Energy Economics 127 (2023) 107104



Contents lists available at ScienceDirect

Energy Economics

journal homepage: www.elsevier.com/locate/eneeco





Which combinations of renewable energy policies work better? Insights from policy text synergies in China

Xing Hu [a,b], Shiwei Yu [a,b,*], Xu Fang [a,b], Marten Ovaere [c,d]

[a] Center for Energy Environmental Management and Decision-making, China University of Geosciences, Wuhan 430074, China
[b] School of Economics and Management, China University of Geosciences, Wuhan 430074, China
[c] Department of Economics, Ghent University, Ghent 9000, Belgium
[d] School of Environment, Yale University, United States of America

ARTICLE INFO

Keywords:
Renewable energy policy
Synergistic effectiveness
Policy text
Policy tool

ABSTRACT

Policy synergy is crucial to achieving unified national strategies in the effective implementation of policy. This study investigates the synergistic effectiveness of China's renewable energy (RE) policy tools by analyzing 7920 RE policy texts issued by provincial governments between 2005 and 2020. By proposing the "Policy Text Quantification-Association Rule Mining-Synergy Effect Test" framework, we explore how the interaction between five policy tools affects Chinese provincial RE development. Results show that provincial policy texts mainly use targets & planning and industrial development with lower stringency than the other three tools. More importantly, targets & planning policies interact positively with operations management and industrial development, as they help reduce the uncertainty during concrete implementation in achieving expected targets and plans. On the other hand, industrial development and financial support policies could potentially repeat incentives on specific industries or enterprises, amplify unequal resource allocation and market distortion, failing to generate synergy effects.

1. Introduction

The urgency to address climate change and environmental degradation has spurred an international consensus on the pivotal role of renewable energy (RE) in curbing emissions and transitioning towards sustainable energy sources. As of 2022, 183 signatories to the Paris Agreement have included RE components in their nationally determined contributions, with 143 having set quantified targets (IRENA, 2022). As a pioneer of RE development, China has been the country with the fastest growth rate of RE installed capacity globally since 2010. In 2022, the RE installed capacity and generation reached 1213 GW and 2700 TWh, respectively (NEA, 2023). However, in 2021, China's RE consumption only accounted for 14.2% of primary energy consumption (NEA, 2022), which is lower than 46.2% in Brazil, 40.3% in Denmark, 29.9% in Canada, and 19.5% in Germany (BP, 2022). There is still a considerable gap between the share of RE consumption and China's "dual carbon" goal, where RE accounts for 25% in 2030 and more than 80% in 2060. In addition, renewable power curtailment still exists due to the incoordination between RE installed capacity, power transmission, and consumption (Yu et al., 2022). Therefore, to achieve its low-carbon energy transition and carbon neutrality goal, China still needs to make more efforts to develop RE.

Implementing refined and synergetic RE policies is an important measure to promote the further development of RE in China. Since the *Renewable Energy Law* in 2006, approximately 8000 RE-related policy documents have been released by China. Implementing these policies has undoubtedly enabled the rapid development of RE in China. However, these policies also lack pertinence and insufficient timeliness, especially a lack of synergy. Policy synergy refers to the harmonious state of public policy, wherein deficiencies, redundancies, and inconsistencies among policy elements are minimized, and cooperation is maximized (Peters, 2015). It becomes a critical consideration during policy design, promulgation, and implementation (Hamada, 1976; Oudiz and Sachs, 1985). The non-synergistic RE policy brings confusion during implementation, reduces the normativity of the policy, and makes it difficult to give full play to its role in promoting the development of RE. It may also lead to renewable power curtailment and overcapacity (Yu et al., 2021). Policy tool serves as the mean to achieve intended policy goals. Successful synergy among policy tools and effective policy combinations can complement weaknesses and promote

* Corresponding author at: Center for Energy Environmental Management and Decision-making, China University of Geosciences, Wuhan 430074, China.
E-mail addresses: yusw@cug.edu.cn, ysw81993@sina.com (S. Yu).

https://doi.org/10.1016/j.eneco.2023.107104
Received 14 July 2023; Received in revised form 14 September 2023; Accepted 3 October 2023
Available online 12 October 2023
0140-9883/© 2023 Elsevier B.V. All rights reserved.

strengths of different policy instruments, while counterproductive combinations might cause policy redundancy and conflicts (Kwon, 2018). Therefore, to further promote the development of RE in China, it is necessary to investigate the synergistic effectiveness of China's RE policy tools and identify the combinations that minimize the risk of policy conflicts and failures among multiple policy tools.

Policy synergy has been extensively studied across various policy domains, including monetary and financial (Carrillo et al., 2021), science and technology (Pang et al., 2020), land use (Lambin et al., 2014) and environmental and climate policies (Li and Ye, 2021; Viguié and Hallegatte, 2012). Relevant studies around policy synergy have primarily focused on its necessity (Canzoneri et al., 2005), challenges (Ostry and Ghosh, 2016; Peters, 2018), comprehensive evaluation (Hughes et al., 2013; Metcalfe, 1994), and impact on environmental as well as economic performances (Zhang et al., 2023a).

However, previous literature on synergy among different policy tools primarily focused on theoretical analysis or modeling the synergy between two or several specific policy measures. Three types of policy interactions can occur depending on how multiple policy tools are combined and implemented (van den Bergh et al., 2021). The first type is a positive interaction, where the combination of policy tools generates a synergy effect, resulting in greater effectiveness than using any one tool in isolation. Combining multiple tools is popular and common in most circumstances because it can simultaneously address multiple externalities and various market failure issues (Gunningham and Sinclair, 1999). Lema and Ruby (2007) reviewed the development and reforms of China's wind energy policies, including the initial, incremental, and coordinated phases. They found that bureaucratic centralization can help reduce policy conflicts while coordinated market regulations and incentives led to significant progress in wind power installations and cost reductions. Additionally, Kwon (2018) believed that a policy mix of RPS and auctions could effectively mitigate market risk problems in South Korea, such as the failure to sell renewable electricity or price volatility.

Negative interaction, the second type, occurs when potential trade-offs and negative externalities arise from the simultaneous implementation of different policy tools, leading to policy conflicts or synergy failure. The inconsistencies in the content of different tools prevent implementers from following uniform standards and can sometimes result in unintended consequences (Nilsson et al., 2012). Also, combining multiple policy tools with similar strengths and shortcomings may repeatedly address the same market failure, leading to policy redundancy (Fankhauser et al., 2010). Furthermore, conflicting interests among policy issuers can also lead to limited interactions among administrative departments related to relevant tools, thereby hindering the actual effectiveness of policy implementation (Canzoneri and Gray, 1985). Some studies found that combining RE with climate policy instruments can have counterproductive effects. For example, Linares et al. (2008) argued that double-counting incentives or geographical overlap between the Emissions Trading Scheme and RES-specific measures might occur. Based on the partial equilibrium model, Tu and Mo (2017) indicated that renewable subsidies may induce the collapse of CO_2 prices. A negative synergy effect was also found between RE quotas or subsidies to the carbon market(van den Bergh et al., 2021). Additionally, del Río (2014) supported the need for policy coordination but found a limited role of policy synergy due to the trade-offs at different criteria when evaluating the challenges of policy energy between targets, tools, and administrative levels with the case of RE support and emissions trading schemes.

The third interaction, zero interaction, refers to policy tools not influencing each other's effects when combined. In this case, combining policy tools merely stacks simultaneous implementations (Fankhauser et al., 2010).

Regarding the synergy of RE policies, existing studies widely discuss synergies cross borders, regions, or administrative institutions but fail to consider them from the perspective of policy texts fully. Bistline et al. (2020) assessed the impact of renewable policy synergy on the electric sector across North American countries and found that synergy can lower costs by up to 20%. By simulating and comparing the synergy schemes that Renewable Energy Certificates (RECs) can or cannot be traded between the U.S., Canada, and Mexico, Avraam et al. (2021) concluded that greater regional synergy would mitigate the impact of RPS on natural gas production, prices, and trade. Zha et al. (2023) utilized a continuous difference-in-differences regression model to evaluate the individual effect and synergistic effect of four renewable energy policies on carbon reduction of China's power industry. While other scholars have also explored the synergy effect among different administrative institutions (Lema and Ruby, 2007), policy goals (Friesenbichler, 2016) and etc., few have focused on analyzing policy documents. On the one hand, the ignorance of their synergies might induce controversy in implementation and even worse consequences for policy effectiveness. On the other hand, existing approaches that simulate synergy effects under various political scenarios have limited applicability in provincial policy texts. Based on a policy text analysis, Chong et al. (2023) quantitatively investigated the policy synergies of 117 photovoltaic power generation policies in China from the perspectives of cross departments, cross goals, and cross measures. However, they overlooked the role of secondary policies,[1] which might result in biased estimations of the synergistic effect and flawed conclusions to some extent.

To address this, the provincial stringency measurement of policy texts and the construction of a new research framework is needed, which could overcome the current limitations on empirical analysis of policy effects and synergy at the regional level. The present study first quantifies the policy stringency of five tools based on 7920 provincial RE policy texts. To reduce the overlap and highlight the critical significance of policies, 483 policies are considered primary policy texts, whereas 7437 are secondary policies. And then, the interaction terms of representative policy combinations with the strongest associations are picked out to introduce into the synergy effect test model and identify their synergistic effectiveness.

This study contributes to the literature from the following three perspectives. a) A rating scale of five RE policy tools (targets & planning, industrial development, financial support, price regulation, and operation management) at the province level is developed from the policy strictness and rigor. It provides a base on quantifying policy stringency and observing characteristics and differences in regional renewable energy policy tool synergies and effects. b) An integrated framework of the "Policy text quantification-Association Rule Mining-Synergy effect Test" is designed to investigate the synergistic effect of RE development and identify effective policy combinations. c) The roles of secondary policies are considered by setting different scenarios they might act when quantified to avoid the over- or underestimated of the policy effect.

2. Methodology and data

2.1. Framework

The existing quantitative research paradigms concerning policy synergy and its effectiveness can be categorized into four distinctive avenues. (a) Comprehensive index development. One approach has endeavored to undertake an multicriteria or multiple attribute analysis and construct an all-encompassing evaluation index for policy synergy

[1] Secondary policies are not specifically designed for RE development, but do state requirements for RE. For example, issues related to RE are also addressed in the *Energy Conservation Law*, environmental and climate protection related laws and plans or documents as a type of clean energy (Liu, 2019. China's renewable energy law and policy: A critical review. Renewable and Sustainable Energy Reviews, 99, 212–219. https://doi.org/10.1016/j.rser.2018.10.007.)

degree(Liu and Wang, 2022; Shrestha and Dhakal, 2019; Viguié and Hallegatte, 2012); (b) Regression analytics. An alternate path involves conducting rigorous regression analyses, wherein the quantified synergy index is regressed against various policy-related outcome variables (Li et al., 2023; Lu et al., 2022). This analytical method permits the nuanced estimation of policy synergy's elastic influence on the outcomes of implementation; (c) Interaction dynamics exploration. Some studies focus on the quantification of policies and the subsequent exploration of synergy through observing the interplay among diverse policy variables (Dong, 2012; Pang et al., 2020; Zha et al., 2023); (4) Scenario-based comparison. Lastly, a branch of research concentrates on the comparative analysis of simulation outcomes between scenarios where policies are implemented individually and implemented with combination or coordination (Abrell and Rausch, 2016; Liu et al., 2023; Mulder and Zeng, 2018).

Realizing that policy text is the carrier conveying the core content of public policy, several studies have incorporated textual data into assessing policy synergy and performance across horizontal, vertical, and temporal dimensions in the first two paradigms. Nonetheless, these studies often calculate the synergy degree among different policy tools by directly multiplying their scores. This approach inevitably establishes a proportional relationship between the synergy degree and the diversity of tools, potentially ignoring the possible negative interaction that may arise from combining multiple tools. The Scenario-based comparison tends to center primarily on the institution itself, often overlooking critical details within the policy text and the role of secondary policies. As for the third research paradigm, most studies rely on adopting general proxies for policy indicators, such as binary dummy variables, policy counts or specialized indicators designed to capture key information. However, these studies also fail to consider the strictness of policy release and the rigor of policy content.

Therefore, a research framework of "Policy Text Quantification-Association Rule Mining-Synergy Effect Test", as depicted in Fig. 1, is constructed to investigate the synergistic effectiveness of provincial RE policy texts in China. Specifically, it seeks to answer questions such as the presence of positive, negative, or zero interaction effects among the policy tools within these texts and which combinations of policy tools are effective in promoting RE development. The framework not only addresses above limitations, but also allows us to identify key combinations of policy tools.

The initial step of our analysis involves quantifying the policy stringency of 30 provinces in China using 7920 provincial RE policy texts. This is accomplished by considering both the strictness and rigor of policy documents. We test for synergy effects among policy tools using a two-way fixed effects model with a panel of 30 provinces over 16 years. However, as 26 potential combinations of the five policy tools exist, incorporating too many explanatory variables may result in a loss of degrees of freedom, thus diminishing the reliability of the estimated results. Testing the synergy effect of all possible combinations simultaneously within the same regression equation is thus not feasible. As a quantitative and fine-grained set of rules, association rules allow decision-makers to find the relationships and characteristics of co-occurring items from large databases expeditiously. By setting minimum support and confidence thresholds, association rules mining can identify policy tool combinations with strong correlations whose components tend to generate potential interactions. We, therefore, first use Association Rule Mining (ARM) to identify the representative policy tool interactions to be included in the econometric model and avoid subjective influence.

The proposed research framework offers two main advantages. Firstly, quantitative analyses of renewable policy texts can help monitor and track the RE policy evolution and differences among provinces, providing a cornerstone for empirically evaluating the effectiveness of RE policies and their synergies at the provincial level. More importantly, comparing and benchmarking policy stringency facilitate local governments to learn from each other in adopting RE policies and improving the RE policy system. Secondly, by integrating text analysis, data mining, and an econometric model, the proposed framework can provide policymakers with a more holistic and evidence-based perspective on policy issues. It enables the extraction of valuable information from unstructured data and avoids potential biases arising from the subjective selection of various policy combinations.

2.2. Quantification of policy texts

2.2.1. Policy collection and classification of policy tools

We utilized web crawler technology to gather policy data from the PKULaw database[2] and official government websites of 30 Chinese provinces. As China's hydropower industry developed earlier and has already reached a relatively mature stage compared to other RE sources, the study mainly focuses on exploring policy synergies of wind, solar, and biomass. The keywords "wind power", "wind energy", "photovoltaic", "solar energy", "biomass", "renewable energy", and "new energy" were used to retrieve China's local policy texts from regulations, orders, normative documents, and working documents at the provincial level from 2005 to 2020. We recorded the essential details of each policy text, encompassing their titles, relevant descriptions related to renewable energy, the form of policy documents, the issuing department, the date of issuance, the date of expiration or modification, and the source or corresponding uniform resource locator (URL). After eliminating the duplicate and irrelevant contents, the final dataset of 7920 RE policy texts was retained as the research object.

The classification of policy tools by scholars displays considerable divergence, driven by applying various criteria. In essence, a prevailing trend in most studies focuses on inherent attributes of policy tools (Landry and Varone, 2005; Salamon, 2002), the types of government resources utilized (Hood, 1983), and the approaches through which government provides guidance, serving as pivotal criteria for classification (McDonnell and Elmore, 1987; Schneider and Ingram, 1990; Vedung, 1998). Among them, based on sources of motivation driving technological change and innovation, Costantini et al. (2015), Ren and Albrecht (2023), and other scholars categorized public policies into demand-pull and technology-push types. Howlett and Ramesh (1995) proposed a classification framework with three categories (mandatory, voluntary, and hybrid types) according to the policy tools' compulsion level. According to the government's way of providing guidance, another categorization scheme esteemed by scholars is to classify government intervention into command-and-control and market-based (Beiser-McGrath et al., 2023; Tang et al., 2020; Tuladhar et al., 2014). Command-and-control policies address explicit mandates or restrictions governing public activities, while market-based policies are believed to result in "cost-effective" allocations and provide continuous dynamic incentives (Jaffe and Stavins, 1995).

This study subdivided and elaborated upon the classification within the framework consisting of command-and-control and market-based policies to uncover a wider array of intricate policy interactions and synergies. In addition, a manual review of the textual content of Chinese provincial renewable energy policies reveals that, due to China's institutional context and the specific features of its RE development stage, there are also policy tools that exhibit a blend of characteristics from both command-and-control and market-based policies. Therefore, we categorized the policy tools related to RE into five categories: Target & Planning, Industrial Development, Financial Support, Price Regulation, and Operations Management, as detailed in Table 1. Among them, Target & Planning and Operations Management policies belong to command-and-control tools. In contrast, Industrial Development and

[2] PKULaw database refers to the Peking University Legal Information Database, which compiles public policies that have been promulgated in mainland China since 1949. The database can be accessed through its website at https://www.pkulaw.com/.

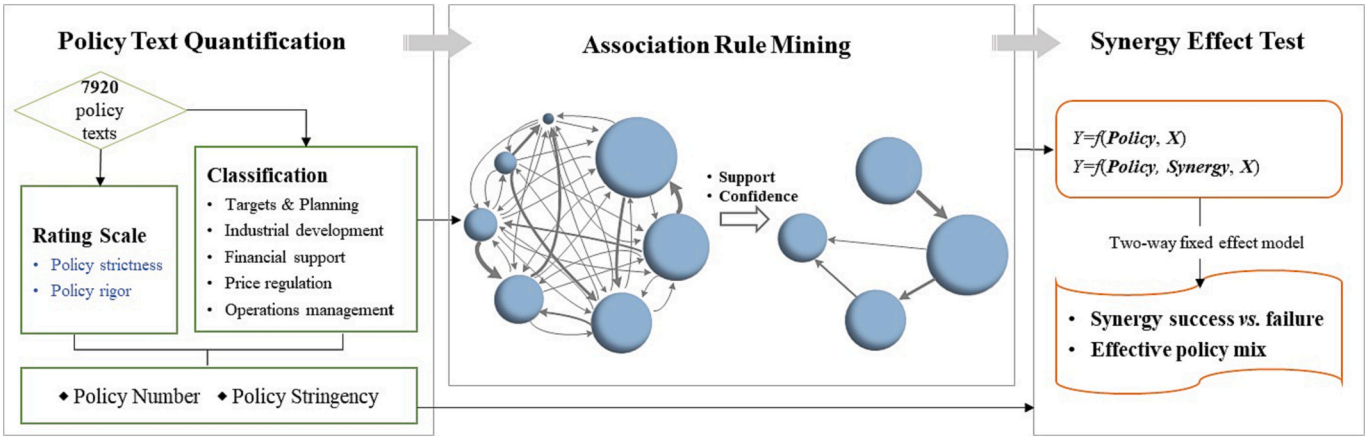


Fig. 1. Research framework.

Table 1
The classification of policy tools.

Policy types	Connotation
Targets & Planning	To set specific goals or deployment plans for RE development and utilization
Industrial Development	Strategic initiatives to support RE-related product (eg. solar panel, photovoltaic cell) and equipment (eg. wind turbine blade, gearboxes and etc.) manufacturing industries. For example, training of professional and technical personnel, improvement of technical standards, support for technological breakthroughs, and public exhibitions of novel technologies.
Financial Support	Fiscal and financial support for RE-related projects through investment and electricity price subsidies, green finance, tax incentives, etc.
Price Regulation	The measures or adjustments of RE electricity price to improve the pricing mechanism
Operations Management	The regulatory and operational management measures formulated during the process of RE resources development and grid connection or the production and operation of RE products and their supporting infrastructure, including but not limited to rules of vetting, approval, records, supervision, and assessment of planned RE projects construction and systematic management measures of enterprise investment as well as operations.

Financial Support are considered market-based ones. Price Regulation policies, on the other hand, have both mandatory and market attributes.

Further, we label the policy texts whose title refers explicitly to the above keywords as "primary policies" and the rest as "secondary policies". Besides, policy texts with two or more tools are marked as "multiple policies".

2.2.2. *Measurement of policy stringency*

Policy stringency refers to the level to which policies are strict and rigorous in their implementation and enforcement. Botta and Koźluk (2014) viewed environmental policy stringency as the explicit or implicit cost imposed on polluting or other environmentally harmful behavior. Hille and Shahbaz (2019) regarded stringency as the cumulative effect of politically determined policies restricting emissions. Meanwhile, Zhang et al. (2022) considered "policy stringency" and "policy intensity" as synonyms, defining policy intensity as the extent of stringency in the delivery and output of a policy document.

The most common approaches for measuring policy stringency are compliance-cost and specific proxies directly relevant to the targeted issues. To estimate the compliance-costs of environmental policies, numerous studies have implemented a shadow price approach, sector-specific energy prices, or environmental R&D expenditures as the pollution abatement efforts (Coggins and Swinton, 1996; Hille, 2018; Sato and Dechezleprêtre, 2015). Regarding specific proxies, some researches tended to quantify the stringency by capturing specific information from the policy. For instance, Carley and Miller (2012) and Carley et al. (2018) proposed the RPS stringency score according to the mandate requirement and percentage of renewables and the electricity load of 49 U.S. states. Others have used direct environmental outcomes, such as emissions, pollutant reductions, or energy use (Costantini and Crespi, 2008; Yan et al., 2023). Based on emission indicators, expenditures, and specific information of targets issues, Botta and Koźluk (2014) developed two composite environmental policy stringency indexes to quantify the "cost" of energy sector and "economy wide", respectively, which have been widely adopted in recent environmental research studies (Martínez-Zarzoso et al., 2019; Saussay and Zugravu-Soilita, 2023; Sun et al., 2022). In addition, other scholars such as Brunel and Levinson (2016) and Galeotti et al. (2020) have also built composite indexes to measure policy stringency.

However, the above studies have primarily focused on defining and measuring policy strictness without considering the rigor of policy documents. To fill this gap, we have developed a rating scale for policy strictness and rigor (as shown in Table A1 of **Appendix 1**) better to quantify the policy stringency of China's RE policy tools.[3] Specifically, policy strictness is determined by the form of policy documents and the level of the administrative institution issuing simultaneously (Liao, 2016). On the other hand, policy rigor depends on the clarity, detail, and completeness of text describing each type of policy tool. The higher the policy strictness and rigor of a document, the more stringent it is for policy stakeholders.

We obtain the strictness score (S_k) of policy text $k(k = 1, 2, ...7920)$ using the criteria of Table A1. Considering the timeliness of the policy's release and implementation, it is assigned the same score within the validity period ($t \in [m_k, n_k]$), whereas the score equals 0 in other periods. That is, the strictness of policy text k in year t is as follows:

$$S_{tk} = \begin{cases} S_k & t \in [m_k, n_k] \\ 0 & otherwise \end{cases} \quad (1)$$

The rigor score (R_{kj}) of the policy tool $j(j = 1, 2, \ldots, 5)$ contained in the policy text k is defined in the same way as shown in Eq. (2).

$$R_{tkj} = \begin{cases} R_{kj} & t \in [m_k, n_k] \\ 0 & otherwise \end{cases} \quad (2)$$

Combining the strictness and rigor score, the policy score of policy tool j in year t is given by,

[3] To improve the data quality and validate the credibility of the manually rated scores, we followed the processing of Zhang et al. (2022).

$$RS_{tj} = R_{tkj} \cdot S_{tk} \cdot w \tag{3}$$

where w represents the importance of secondary policies compared to primary policy texts. In the remainder of this paper, we will assume $w =$ 0.25, 0.5, 0.75, and 1. The policy stringency (PS_{itj}) of each policy tool (j = Targets & Planning, Industrial Development, Financial Support, Price Regulation and Operations Management) of province i in year t is then obtained as the sum over the policy score of all policy tools issued by province i (h_i), divided by the number of policy tools (PN) in this province, respectively.

$$PS_{itj} = \left(\sum_{k \in h_i} RS_{tj}\right) \Big/ PN_{itj} \tag{4}$$

2.3. Association rule mining

Association rule mining (ARM) is a powerful data mining technique that extracts interdependencies and associations between various items through frequent itemset analysis. (Agrawal et al., 1993). By exploring the intricate patterns and relationships among co-occurring events, ARM can unravel intriguing insights for decision-makers without being restricted by the number of dependent variables (Han and Fu, 1999; Toivonen, 1996). In recent years, ARM has been widely used in various fields ranging from basket analysis(Ünvan, 2021) to medical diagnosis (Tandan et al., 2021), risk prediction and management (Dehbozorgi et al., 2022; Shang et al., 2021), complex evaluation issues (Antonello et al., 2021; Wang et al., 2018). This study aims to utilize ARM to uncover strong correlations among various policy tools in a database comprising 7920 RE policy texts. Through this endeavor, representative forms of policy combinations are identified.

Let $I = \{I_1, I_2, \ldots, I_n\}$be a set of items and T be the transaction. Each transaction corresponds to a subset of I such that $T \subset I$. All transactions are contained in the database M, namely $M = \{T_1, T_2, \ldots, T_n\}$. Set X and Y as two different subsets of I, namely, itemsets. Thus, an association rule can be expressed as $X \rightarrow Y$, where $X, Y \subset I$ and $X \cap Y = \varnothing$. X is called antecedent and Y is the consequent. The rule$X \rightarrow Y$means that X implies Y a certain probability. This study I includes five items of RE policy tools, that is, $I = \{TP, ID, FS, PR, OM\}$. T refers to the set of policy tools used in each policy text. M indicates the 7920 policy tool combinations of all policy texts.

Analyzing the results of association rule learning typically involves using support and confidence as the two main criteria. The support reflects the frequency with which a rule appears in the dataset, while confidence gauges the strength of the relationship between the antecedent and consequent of the rule. As shown in Eq. (5), the $Support(X \rightarrow Y)$ is calculated as the frequency of the transactions that X and Yoccur simultaneously in the database. As for confidence, it is defined as the conditional probability of the consequent Y given the antecedent X, as presented in Eq. (5).

$$Support(X \rightarrow Y) = \frac{n(X \cap Y)}{|M|} \tag{5}$$

$$Confidence(X \rightarrow Y) = \frac{Support(X \rightarrow Y)}{Support(X)} = \frac{n(X \cap Y)}{n(X)} \tag{6}$$

Where $|M|$ is the number of transactions, $n(X \cap Y)$is the number of transactions that contain both X and Y simultaneously. The indicator could filter all relative itemset meeting the specified frequency threshold. According to Saquete et al. (2022), the present study's minimum support threshold is 0.1. The Apriori algorithm and specific implementation process are shown in **Appendix 2**.

2.4. Proposed model

Apart from conducting scenario simulations(Eker and van Daalen, 2015; Hombach and Walther, 2015; Mulder and Zeng, 2018), counterfactual analysis framework(Dong et al., 2021; Du and Takeuchi, 2020) and comprehensive evaluation models(Popiolek and Thais, 2016; Sovacool, 2010), existing studies also frequently investigate the policy effectiveness through regressions involving relevant policy outcome variables with quantitative policy indicators, such as dummy variables or binary variables(Bersalli et al., 2020; Zhao et al., 2016), the duration of policy implementation(Carley, 2009), the number of policies(Hu et al., 2022; Liu et al., 2019), or other specific proxies that reflect important characteristics(Palage et al., 2019; Schmidt et al., 2013). This paradigm can effectively capture the extent of elasticity in the changes of interesting outcome variables concerning policy indicators. Using an ordinary least squares regression, Menz and Vachon (2006) examined the contribution of several wind policies in the U.S. states by employing dummy variables representing policy implementation (0 before, 1 after) and the duration of policy enactment (policy experience), respectively. Yin and Powers (2010) introduced a new consideration of the state-by-state strategy design of RPS stringency as a metric and applied a fixed effects model. Their findings concluded that RPS policy promotes the deployment of renewable energy.

In the present study, we develop a policy stringency index considering policy strictness and rigor simultaneously. The significant and positive role of policy stringency thus indicate that the growth in renewable energy development is associated with the enhancement of government intervention, thereby suggesting the effectiveness of corresponding measures.

We initially propose a model that neglects the potential synergies among policy tools, as depicted in Eq. (7). In recognition of the potential time lag in policy effects, and we introduce lagged forms of policy variables into the model. This strategy aims to mitigate endogeneity bias that may stem from bi-directional causality between policies and RE development. Furthermore, by comparing and analyzing the outcomes of different lag lengths, we can delve into the dynamic changes in individual and synergistic effectiveness, allowing for a deeper comprehension of the complex interplay among policy tools over time.

$$lnRED_{it} = \beta_0 + \beta_1 lnTP_{it-l} + \beta_2 lnID_{it-l} + \beta_3 lnFS_{it-l} + \beta_4 lnPR_{it-l} + \beta_5 lnOM_{it-l} + \delta \mathbf{Control}_{it} + \mu_i + \gamma_t + \varepsilon_{it} \tag{7}$$

Where RED represents the development of RE of province i in period t. TP,ID,FS,PR,OMindicate policy stringency degree of *Targets & Planning*, *Industrial Development*, *Financial Support*, *Price Regulation*, *Operations Management* respectively. l implies the number of lags. **Control** is the vector set consisting of control variables, and δ is the vector of corresponding estimated coefficients. β_0, μ_i, γ_t and ε_{it} are the intercept term, the individual effect, the time effect, and the disturbance term, respectively.

Being constrained by geographical and climatic factors, RE is predominantly applied and transmitted through electricity. Thus, the installed capacity becomes a vital indicator of RE development and deployment as it indicates the maximum capacity of renewable power generators over a particular period, widely employed as the outcome variable in policy studies (Alolo et al., 2020; Dong et al., 2021; Du and Takeuchi, 2020). To focus on the effectiveness of wind, solar, and biomass policies, these renewable sources' per capita generation capacity is utilized to measure the dependent variable.

Prior research has verified that economic growth, the need for an energy transition towards sustainability, and technological advancements have significant contributions to the development of RE (Alam and Murad, 2020; Bourcet, 2020; Marques et al., 2010). Thus, in the current study, three variables-per capita Gross Domestic Product ($PGDP$), Energy Structure (ES), and Technological Innovation (TI)- are considered as the controls. Specifically, energy structure is measured by the proportion of coal consumption to total energy consumption, while per capita RE knowledge stock is adopted as a proxy for technological innovation.

Regarding the policy synergy, we endeavor to discuss the interaction effect among renewable energy policy tool combinations on renewable energy development. To further enhance our understanding of the synergistic impacts of selected combinations of policy tools, we have extended the regression model, as illustrated by Eq. (8).

$$lnRED_{it} = \beta_0 + \beta_1 lnTP_{it-l} + \beta_2 lnID_{it-l} + \beta_3 lnFS_{it-l} + \beta_4 lnPR_{it-l} + \beta_5 lnOM_{it-l} + \beta_6 Synergy_{it-l} + \delta \mathbf{Control}_{it} + \mu_i + \gamma_t + \varepsilon_{it} \quad (8)$$

Where*Synergy*indicates the interaction term between policy tools. In this study, we particularly discuss ARM's representative policy tool combinations.

2.5. Data

The installed capacity data of wind, solar, and biomass generation have been sourced from China Electricity Statistical Yearbook. At the same time, the population size and GDP figures have been collected from China Statistical Yearbook. The China Energy Statistical Yearbook has provided data on coal consumption and total energy consumption, which have been used in calculating energy structure. Referring to the method proposed by Popp et al. (2013), which simultaneously considers technological depreciation and diffusion, the RE knowledge stock has been approximated with the patents granted from the Derwent Innovation patent database (Derwent, 2023). The details of all variables are listed in Table 2, and the descriptive statistics are summarized in Table A2.

3. Provincial characteristics of renewable energy policies

The number of multiple provincial, primary, and comprehensive policies in 2005, 2010, 2015, and 2020 are presented in Fig. 2, respectively.

Overall, there has been a noticeable rise in the number of policy documents issued by provincial governments regarding RE from 2005 to 2020. This upsurge highlights the growing emphasis that local authorities have placed on enhancing RE deployment, utilization, regulation, and management, particularly after the official enactment of the *Renewable Energy Law*. Nonetheless, it should be noted that only a modest fraction of policy documents is exclusively devoted to RE matters. The vast majority of texts are secondary, acting as auxiliary tools to support broader policy objectives such as propelling economic development, advancing industrial transformation, and pollution mitigation. Out of the 7920 policy texts collected in this research, primary policies only accounted for 6.1%. Specifically, the percentages for primary policies are 4.4% in 2005, 5.6% in 2010, 6.5% in 2015, and 7.5% in 2020, respectively.

In the meantime, most policy texts continue to rely on a single policy tool. Over the sample period, the average proportion of policy texts with only one policy tool is 66.8%. Despite the steady increase in the number of policy texts incorporating multiple tools, from a mere 54 in 2005 to 1361 in 2020, their share of the total has followed a slight downward trend, dropping to 30.1% by 2020 after reaching a peak of 37.6% in 2007.

Table 2
Variables and definitions.

	Variable	Definition
Dependent variables	*RED*	RE development level
	TP	Policy stringency of *Targets & Planning*
	ID	Policy stringency of *Industrial Development*
	FS	Policy stringency of *Financial Support*
	PR	Policy stringency of *Price Regulation*
	OM	Policy stringency of *Operations Management*
	Synergy	Set of interaction terms of selected policy tool mixes
Control variables	*PGDP*	Per capita Gross Domestic Product
	ES	Energy structure
	TI	Technological innovation

Figs. 3 and 4 reveal that despite *Targets & Planning and Industrial Development* measures being released more, their degree of policy stringency is relatively low. Out of the total 7920 policy texts, those containing Targets & Planning and Industrial Development constituted 46.8% and 54.2%, respectively. These proportions significantly surpass those involving Financial Support (16.0%), Price Regulation (5.9%), and Operations Management (22.5%). However, taking w = 0.5 as an example, secondary policy text can have half effect of the primary policy text. The policy tool exhibiting the highest level of stringency is Operations Management, with an average of 30 provinces reaching 2.7858 from 2005 to 2020. This is followed by Price Regulation (2.233), Financial Support (2.1822), Targets & Planning (1.9912), and Industrial Development (1.8254).

In addition, the cumulative number of policy tools in most provinces has notably increased since 2010. However, for most provinces, there has been no discernible historical change in each tool's degree of policy stringency. The government's consideration of maintaining authority and credibility may be responsible for this phenomenon. On the one hand, during the research period, the responsibilities of provincial power agencies related to RE did not undergo frequent significant adjustments. On the other hand, the formulation of provincial policy content generally follows the guidelines set forth by the central government as much as possible, and revisions are approached cautiously and gradually rather than accomplished overnight. Consequently, while the number of policy texts and tools has increased with the mounting calls for developing RE, there appears to be little apparent change in the stringency degree of policies over time.

While Zhejiang stands out with the largest quantity of issued policy texts, its policies exhibit a relatively modest stringency. Zhejiang issued 500 policy texts between 2005 and 2020, with an average of 168 tools implemented. However, it does not necessarily suggest a greater importance on RE is attached to Zhejiang than other regions. Regarding the average policy stringency of the five tools, Zhejiang ranks 16th among the 30 provinces with a score of 2.1785. It could be attributed to resource constraints. Most policy texts provide brief mentions without elaborating on specific implementation and development. According to statistics, its average installed capacity was about 4.2 million kW between 2005 and 2020, accounting for only 21.2% of Inner Mongolia.

The three provinces that perform the highest policy stringency are Ningxia, Jiangsu, and Shandong, with average stringency scores of 3.0751, 3.0044, and 2.9674, respectively. Moreover, the average number of policy tools implemented is 163, 235, and 175, ranking 8th, 2nd, and 4th among 30 provinces, respectively. A shared characteristic among these three provinces is their relatively plentiful RE resources, prompting the local governing bodies to formulate more specific and detailed policies concerning RE and thus maintaining policy stringency.

4. Results analysis

4.1. Results of association rules mining

The RE policy tool portfolio information obtained by association rule mining is visualized in Fig. 5. The circle size, the color of the circle, and the line weight represent the count of the policy tool portfolio, the importance of the policy tool combinations (the average confidence level when it is the antecedent of the association) and the strength of the association rule (confidence), respectively.

As depicted in Fig.5(a), policy tools Targets & Planning, Industrial Development, Financial Support, Price Regulation, and Operations Management appear in the second, first, fifth, tenth, and fourth frequency of all 31 policy tool combinations, accounting for 69% of all policy texts. This analysis demonstrates that the Chinese government has employed a diverse range of policy measures to establish a relatively

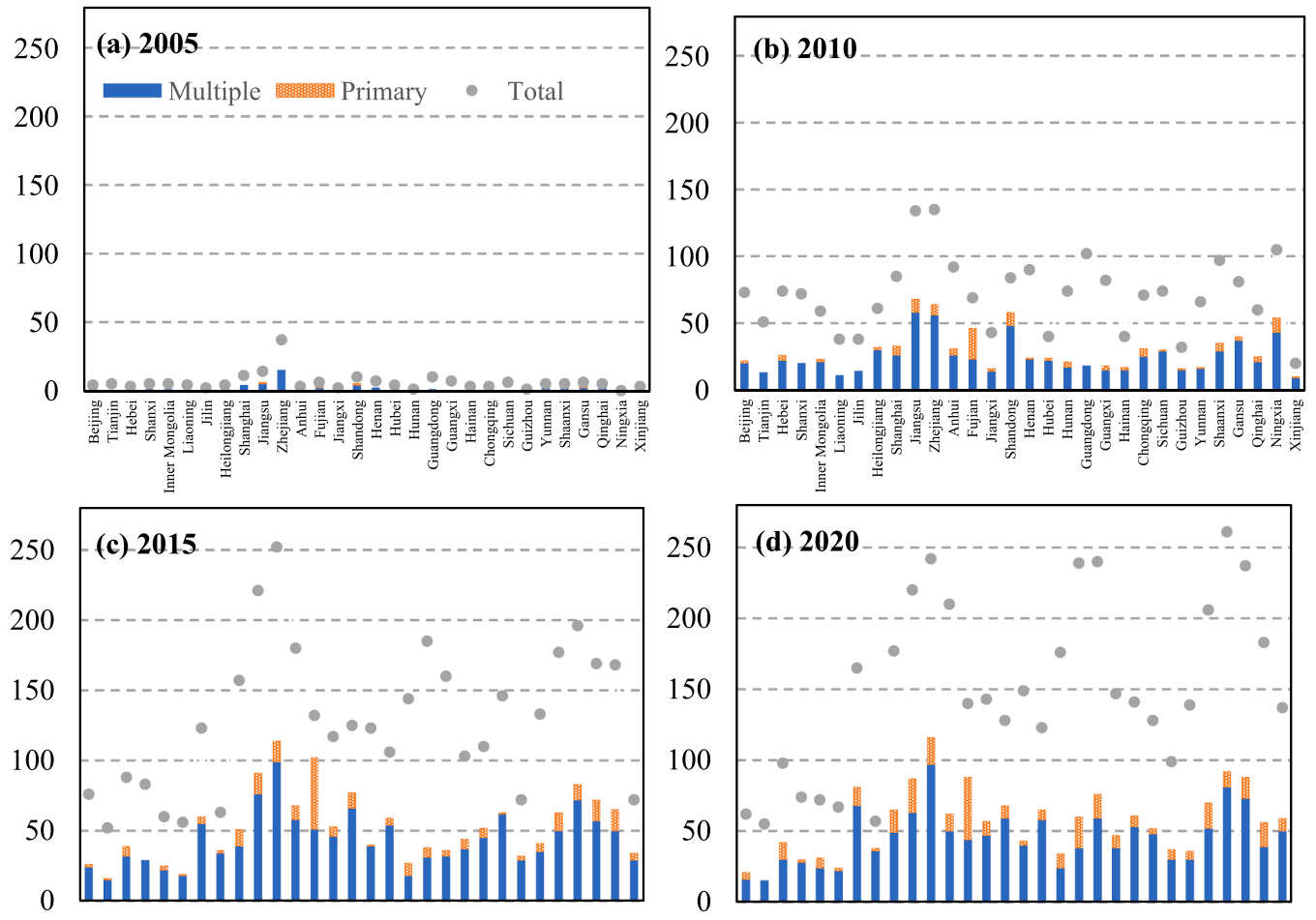


Fig. 2. Number of multiple, primary, and total policy texts.

comprehensive renewable energy policy system, providing robust support for the sustainable growth of renewable energy. More specifically, Industrial Development policies represent the largest proportion at 29.13%, followed by Targets & Planning policy at 22.58%, Operations Management policy at 9.89%, and policies of Financial Support and Price Regulation at 5.86% and 1.44%, respectively. This underscores that, during China's renewable energy development, the government has adopted a target-oriented approach with industrial low-carbon development as the central strategy, complemented by operations management as well as financial support policies. This reflects the "target setting-industry transformation-financial support" logic in the journey towards energy decarbonization (Gielen et al., 2019). Moreover, the deficiency of Price Regulation category is particularly noticeable. It could be because of China's local governments' limited "discourse power" in the context of electricity market pricing and insufficient attention given to renewable energy prices.

Nonetheless, average confidence levels of five sole tools as association antecedents are relatively low, ranking among the lowest tens. It implies that while local governments have gradually recognized the benefits of combining policy tools, most RE policy documents rely on a sole tool and often overlook the potential synergies between them. This tendency may be attributed to local governments' limited attention, resources, and information-processing capacity. The imbalance between confronting complex policy decisions and having restricted resource capacity tends to steer governments towards favoring the utilization of a sole policy tool (Zhang et al., 2023b).

As shown in Fig. 5(b), when comparing the mean confidence levels among various association antecedents, it becomes evident that Price Regulation policy holds the highest position. This policy tool plays a pivotal role in the local renewable energy policy system due to its close connection and collaboration with other policy instruments, as well as its effective impact on the widespread adoption of renewable energy through mechanisms like feed-in tariffs (FIT), tradable green certificates (TGC), and other means (Zha et al., 2023). Following Price Regulation, Financial Support, and Operations Management also assume significant roles, surpassing other policy combinations in the rule-formulation process.

Among the association rules mined, rules "Operations Management→Targets & Planning", "Targets & Planning→Industrial Development", and "Financial Support→Industrial Development" had the strongest association, with confidence levels of 0.4152, 0.3753, and 0.3667, respectively. This suggests that government authorities tend to pair Target & Planning policy with Operations Management policy in their decision-making processes. To illustrate, when aiming to achieve specific renewable energy development objectives during a particular phase, administrators will enact corresponding renewable energy programs and establish operational norms to facilitate policy implementation (Komor and Bazilian, 2005; Zhang et al., 2013). Additionally, the rule of "Targets & Planning→Industrial Development" policy tools signifies that the government optimizes the structure of the energy industry and promotes the effective growth and deployment of renewable energy by crafting detailed targets for the proportion of renewable energy power generation. Furthermore, it is complemented by corresponding financial support when the government introduces industrial development policies. Through the harmonious interaction of these diverse policy tools, a powerful and effective policy "combination punch" is delivered.

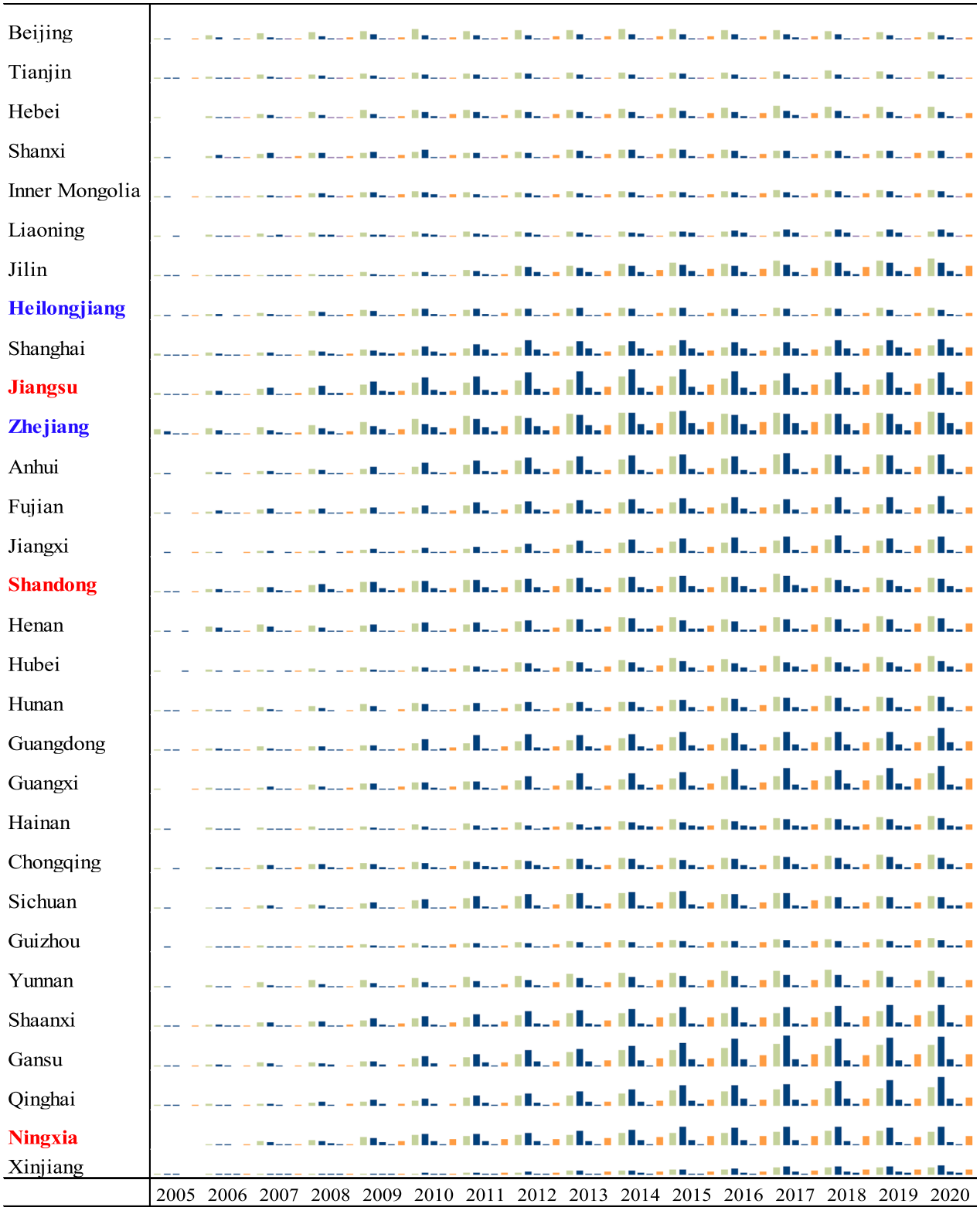


Fig. 3. Numbers of different policy tools of 30 provinces.
Note: The bars in each bar chart, from left to right, are Targets & Planning, Industrial Development, Financial Support, Price Regulation, and Operations Management, respectively.

4.2. Regression results analysis

We substituted the interaction terms of the policy combinations with the three strongest association rules, as identified in the previous section, into Eq. (8) to explore the synergistic effectiveness of specific policy combinations. It is worth mentioning that the policy combination selected through the ARM is more representative, which does not mean that other combinations are unimportant.

Before the regression, several tests were carried out, including Pesaran (2021)'s cross-sectional dependence (CD) test, cross-sectional

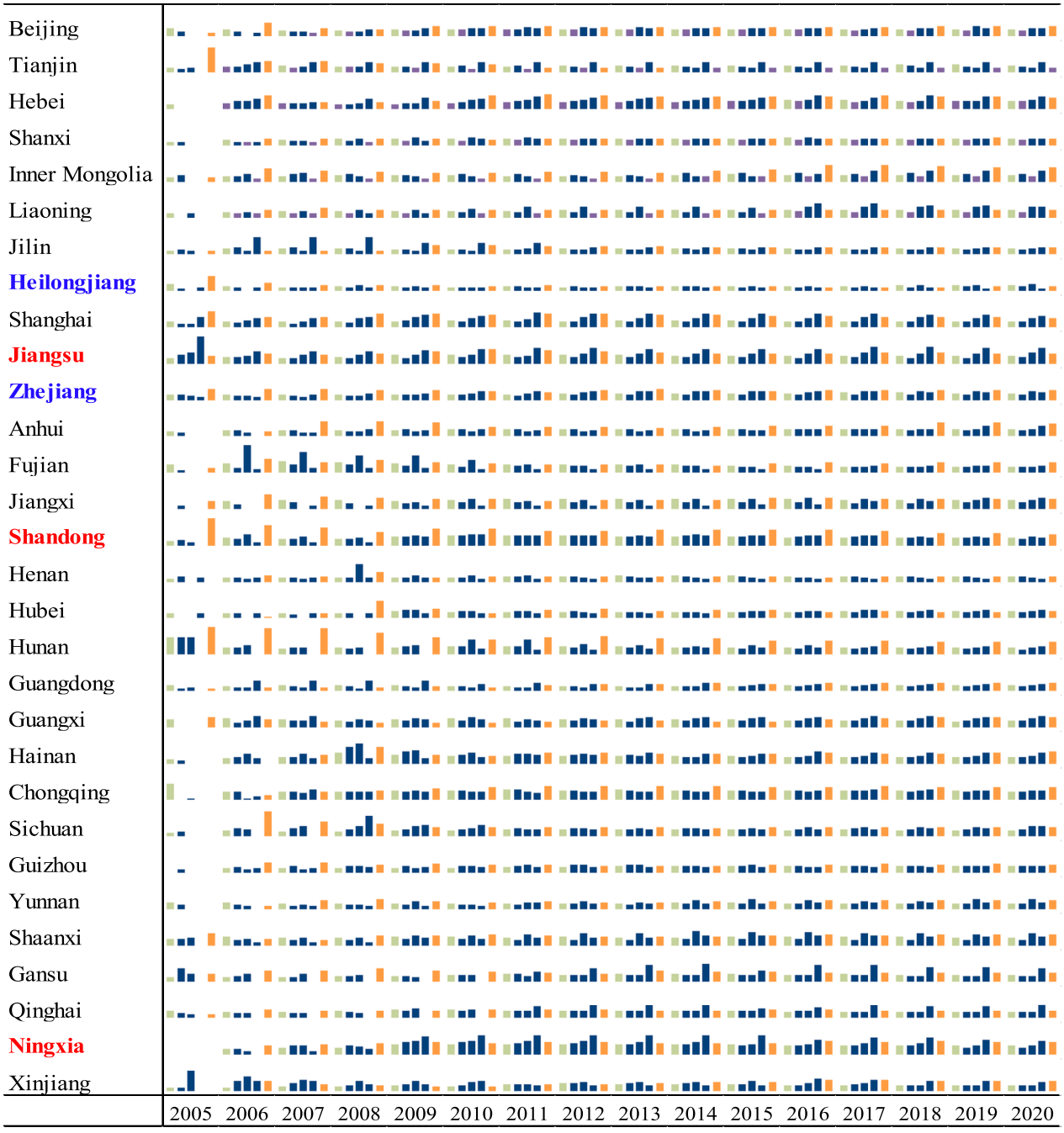


Fig. 4. Policy stringency of different policy tools of 30 provinces ($w = 0.5$).
Note: The bars in each bar chart, from left to right, are Targets & Planning, Industrial Development, Financial Support, Price Regulation, Operations Management, respectively.

Im, Pesaran, and Shin (Im et al., 2003) (CIPS) panel unit root test (Pesaran, 2007) and Kao (1999) panel cointegration test. The results presented in Appendix **Tables A3 and A4** showed that: (1) all variables rejected the null hypothesis that there is no cross-sectional dependence at the 1% significance level; (2) all series were stationary after the first-order difference; (3) at the 1% significance level, all five statistics of Kao (1999) cointegration test indicated that a co-integration relationship exists among the variables included in Eq.(8). Given the presence of cross-sectional dependence, this research applied the fixed effects model with Driscoll-Kraay standard error (Driscoll and Kraay, 1998), which is robust to the panel with cross-sectional dependence, heteroskedasticity and autocorrelation (Hoechle, 2007).

The regression results are presented in **Tables 3** and 4. In each table, Models I to IV display the outcomes when policy interactions are not considered while setting various importance levels for secondary policy texts relative to primary policy texts. Conversely, Model V-VIII incorporates the interactions among policy tools.

Upon comparing the results of Table 3 (one-period lag of policies) and Table 4 (two-period lag of policies), it is evident that the main effects of the five policy tools remain unchanged in terms of their sign and statistical significance. In particular, the main effects of lagged stringency degree of Targets & Planning and Operations Management on the development of RE are observed as significant and positive at a significant level of 5%. Furthermore, the interaction term of Targets & Planning and Operations Management indicates a consistently positive effect, both in the first and second lag periods. This implies a synergy

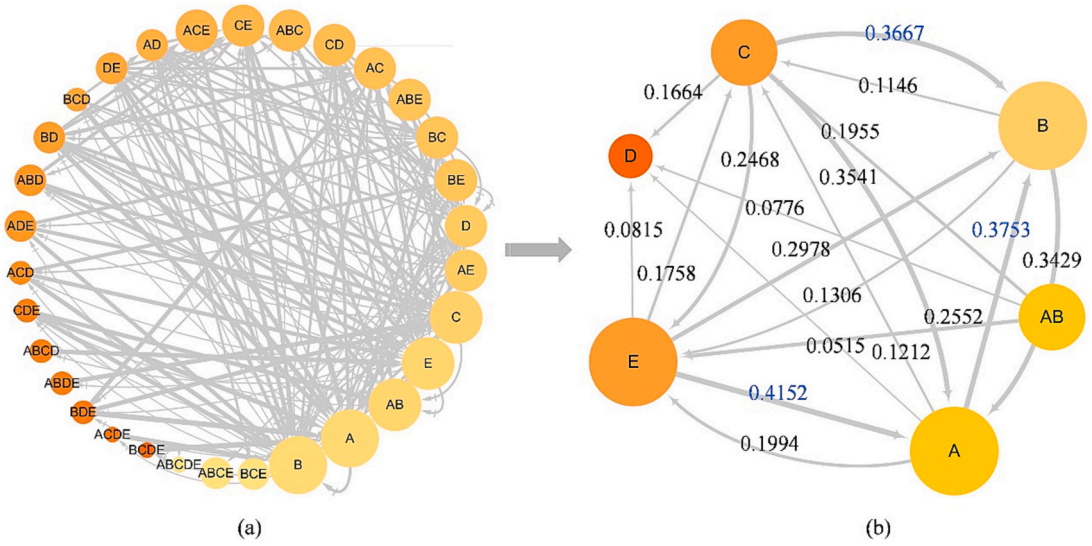


Fig. 5. All possible association rules (a) and rules from frequent itemsets (b) of RE policy tools.
Note: A, B, C, D, and E represent Targets & Planning, Industrial Development, Financial Support, Price Regulation, and Operations Management respectively. The confidence of the three strictest rules is marked in blue.

Table 3
Regression results of representative policy tools synergy on RE development ($l = 1$).

	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
	$w = 0.25$	$w = 0.50$	$w = 0.75$	$w = 1$	$w = 0.25$	$w = 0.50$	$w = 0.75$	$w = 1$
$\ln TP_{t-l}$	0.059***	0.061***	0.062***	0.061***	0.355***	0.345***	0.337***	0.328***
	(0.018)	(0.015)	(0.014)	(0.012)	(0.073)	(0.065)	(0.061)	(0.060)
$\ln ID_{t-l}$	−0.089***	−0.077**	−0.071**	−0.068**	−0.244***	−0.209***	−0.194***	−0.187***
	(0.030)	(0.029)	(0.028)	(0.028)	(0.034)	(0.033)	(0.032)	(0.032)
$\ln FS_{t-l}$	0.027	0.025	0.024	0.023	−0.006	0.005	0.010	0.012
	(0.034)	(0.034)	(0.034)	(0.034)	(0.022)	(0.022)	(0.022)	(0.022)
$\ln PR_{t-l}$	0.136***	0.136***	0.135***	0.134***	0.141***	0.140***	0.139***	0.138***
	(0.046)	(0.045)	(0.045)	(0.044)	(0.046)	(0.045)	(0.045)	(0.044)
$\ln OM_{t-l}$	0.071**	0.069**	0.067**	0.066**	0.130***	0.113***	0.104***	0.097***
	(0.028)	(0.026)	(0.025)	(0.024)	(0.044)	(0.039)	(0.036)	(0.035)
$(\ln TP \times \ln OM)_{t-l}$					0.017**	0.017**	0.017***	0.017***
					(0.007)	(0.006)	(0.006)	(0.006)
$(\ln TP \times \ln ID)_{t-l}$					0.015*	0.014*	0.013*	0.013*
					(0.008)	(0.007)	(0.007)	(0.007)
$(\ln ID \times \ln FS)_{t-l}$					−0.014***	−0.012***	−0.011***	−0.011***
					(0.001)	(0.001)	(0.001)	(0.001)
$\ln PGDP$	13.156***	13.057***	13.006***	12.975***	12.839***	12.696***	12.621***	12.574***
	(2.056)	(2.068)	(2.072)	(2.074)	(2.338)	(2.349)	(2.354)	(2.354)
$\ln ES$	−3.097**	−3.078**	−3.069**	−3.062**	−3.213**	−3.179**	−3.164**	−3.152**
	(1.479)	(1.477)	(1.476)	(1.475)	(1.548)	(1.537)	(1.532)	(1.531)
$\ln TI$	1.707	1.711	1.714	1.713	1.943	1.978	1.993	1.997
	(1.446)	(1.441)	(1.438)	(1.437)	(1.372)	(1.357)	(1.349)	(1.346)
Constant	3.449	3.420	3.390	3.342	5.785	5.809	5.783	5.703
	(4.008)	(4.016)	(4.018)	(4.022)	(3.800)	(3.800)	(3.802)	(3.807)
Individual effect	YES	YES	YES	YES	YES	YES	YES	YES
Time effect	YES	YES	YES	YES	YES	YES	YES	YES
N	450	450	450	450	450	450	450	450
F statistics	288.184***	151.377***	124.183***	106.869***	1.18e+06***	4.98e+06***	9.65e+08***	6.17e+08***
R^2	0.644	0.645	0.645	0.645	0.651	0.651	0.651	0.651

Note: ***, **, and * represent statistical significance at the level of 1%, 5%, and 10%, respectively. Driscoll-Kraay standard errors are in parentheses. w represents the importance of secondary policies compared to primary policy texts.

effect when combining Targets & Planning and Operations Management policies. Well-defined targets and rigorous plans can furnish guidance, foster rational allocation of government resources, and enhance RE development and utilization in a highly effective manner (Hu et al., 2022). While Targets & Planning tools are valuable for establishing requirements in the pre-planning stage, they tend to be inflexible and overlook the importance of subsequent policy implementation. The real world's complexity and uncertainty often hinder realizing intended outcomes and wise trade-offs (Spillias et al., 2020).

Regarding Operations Management, its mandatory attribute offers the opportunity for creating a level playing field for RE development (Ambec et al., 2013). By placing restrictions on the behavior and choices of policy audiences in a strong and binding mechanism, Operations Management policies significantly reduce the risk and uncertainty of policy implementation. Therefore, Operations Management policies could effectively compensate for the limitations of the late

Table 4
Regression results of representative policy tools synergy on RE development ($l = 2$).

	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
	$w = 0.25$	$w = 0.50$	$w = 0.75$	$w = 1$	$w = 0.25$	$w = 0.50$	$w = 0.75$	$w = 1$
$\ln TP_{t-l}$	0.091***	0.092***	0.092***	0.091***	0.399***	0.389***	0.381***	0.372***
	(0.013)	(0.012)	(0.012)	(0.012)	(0.023)	(0.026)	(0.027)	(0.028)
$\ln ID_{t-l}$	−0.221***	−0.207***	−0.200***	−0.195***	−0.190***	−0.170***	−0.163***	−0.159***
	(0.020)	(0.019)	(0.019)	(0.019)	(0.039)	(0.038)	(0.038)	(0.038)
$\ln FS_{t-l}$	0.021	0.018	0.016	0.015	0.019	0.019	0.019	0.017
	(0.023)	(0.023)	(0.024)	(0.024)	(0.022)	(0.024)	(0.024)	(0.024)
$\ln PR_{t-l}$	0.062**	0.061**	0.060**	0.059**	0.060*	0.058*	0.058*	0.057*
	(0.029)	(0.029)	(0.028)	(0.028)	(0.030)	(0.030)	(0.029)	(0.029)
$\ln OM_{t-l}$	0.092***	0.089***	0.087***	0.086***	0.125***	0.110***	0.101***	0.095***
	(0.025)	(0.023)	(0.022)	(0.021)	(0.033)	(0.029)	(0.027)	(0.027)
$(\ln TP \times \ln OM)_{t-l}$					0.017***	0.017***	0.017***	0.017***
					(0.006)	(0.006)	(0.006)	(0.005)
$(\ln TP \times \ln ID)_{t-l}$					0.014	0.013	0.012	0.012
					(0.008)	(0.008)	(0.008)	(0.007)
$(\ln ID \times \ln FS)_{t-l}$					−0.001	0.000	0.000	0.001
					(0.002)	(0.002)	(0.002)	(0.002)
$\ln PGDP$	13.963***	13.957***	13.947***	13.941***	13.207***	13.084***	13.017***	12.982***
	(3.041)	(3.075)	(3.090)	(3.099)	(3.527)	(3.576)	(3.599)	(3.612)
$\ln ES$	−2.513*	−2.506*	−2.503*	−2.500*	−2.539*	−2.528*	−2.523*	−2.518*
	(1.293)	(1.295)	(1.297)	(1.298)	(1.301)	(1.296)	(1.294)	(1.296)
$\ln TI$	2.816**	2.817**	2.818**	2.817**	3.181***	3.224***	3.244***	3.249***
	(1.216)	(1.219)	(1.220)	(1.221)	(1.008)	(1.000)	(0.995)	(0.994)
Constant	−11.882**	−11.890**	−11.888**	−11.899**	−9.807	−9.743	−9.737	−9.777
	(5.397)	(5.437)	(5.453)	(5.460)	(6.624)	(6.737)	(6.791)	(6.821)
Individual effect	YES	YES	YES	YES	YES	YES	YES	YES
Time effect	YES	YES	YES	YES	YES	YES	YES	YES
N	420	420	420	420	420	420	420	420
F statistic	3.08e+05***	3.19e+05***	3.27e+05***	3.18e+05***	3.32e+06***	2.01e+06***	1.67e+06***	1.39e+06***
R^2	0.624	0.623	0.623	0.623	0.630	0.631	0.631	0.631

Note: ***, **, and * represent statistical significance at the level of 1%, 5%, and 10%, respectively. Driscoll-Kraay standard errors are in parentheses. w represents the importance of secondary policies compared to primary policy texts.

implementation of Targets & Planning policies.

As shown in Table 3, lagged $\ln TP \times \ln ID$ is significantly and positively related to RE development, but lagged $\ln FS \times \ln ID$ has the opposite effect. The former finding suggests the positive interactions between command-and-control and market-based policies, which contrasts Tuladhar et al. (2014) 's opinion. The distinct features and strengths of Target & Planning and Industrial Development policies can explain this observation. Industrial Development policies can catalyze job opportunities and provide technical and talent support (Rodrik, 2015), facilitate the implementation of Targets & Planning, and yield a synergy effect to a certain extent. Nonetheless, stricter, and more detailed industrial development policies face inherent risks of unreasonable intervention and the concentration of resources on leading RE enterprises, impeding the marketization development of RE and explaining the adverse role of industrial development policies. Targets & Planning, which clarifies the demand, target, and direction, can help to alleviate the negative effect, while combining Industrial Development and Financial Support may worsen the problem of disparate resource allocation and market distortion. Additionally, the significance of the interactions of Industrial Development with Targets & Planning and Financial Support diminishes when lagged second-period variables are introduced. It underscores that timely and valid public policy is essential for sustaining long-term efficacy of policy combinations and synergies.

The larger the relative importance of secondary policy text to primary policy text (w), the weaker the policy effect would have. From Model V to Model VIII of Tables 3 and 4, it can be drawn that the absolute values of their estimated coefficients almost decrease as the w increases, regardless of the primary or interaction effect. The possible reason may be twofold. On the one hand, secondary policy texts designed to address other policy objectives may contain vague and simplistic descriptions about RE with low rigor score. The redundancy and wastage of policy resources could easily occur when secondary policy texts hold closer importance, resulting in a diminished effect of policy tools. On the other hand, multiple stakeholder sectors might be involved in achieving various main policy goals of the secondary policy texts, increasing the likelihood of encountering conflicts or frictions. It could also induce unexpected consequences for the advancement of RE.

4.3. Robustness check

4.3.1. Alternative measurements of variables

This section replaces RE's per capita installed capacity with a cumulative installed capacity to measure RE development level. The results are presented in Table A5 of the Appendix 5. Additionally, to mitigate impact of potential endogenous problem arising from bidirectional causality between control variables and RE development, we incorporate lagged forms of control variables for a robustness check in Table A6. Upon comparing these findings with those in **Tables 3** and **4**, we observed that Target & Planning policies interact positively with operations management and industrial development policies, while the interaction between industrial development and financial support policies are significantly negative. Besides, the policy effects get weaker with time and with the increase in important level of secondary policy relative to primary policy. These results consistently indicate the robustness of the present study.

4.3.2. Changing sample period

In light of the widespread restrictions imposed on social production activities during the Covid-19 pandemic since 2020, a substantial deviation in data patterns may arise. To mitigate potential anomalies caused by these unforeseen circumstances, this section opts to change the sample period from 2005 – 2020 to 2005 – 2019 for conducting a robustness analysis. The regression results, presented in Table A7 of **Appendix 5**, reaffirm the robustness of our original findings, further consolidating the validity and reliability of the conclusions across different timeframes.

4.3.3. Centering interaction terms

The interaction terms in this study were generated by multiplying two policy stringency indictors, which may raise concerns regarding multicolinearity and potentially masking the true contribution of policy synergy. To address above issues and ensure the resilience as well as stability of the conclusions, the interaction terms were obtained with centered variables (Robinson and Schumacker, 2009) in this section to assess the robustness. The coefficients, sign and significance presented in Table A8 continue to support our main findings.

5. Conclusions and policy implications

5.1. Conclusions

Utilizing 7920 RE policy texts by provincial governments in China, this study constructs a research framework of "Policy Text Quantification-Association Rule Mining-Synergy Effect Test" to empirically analyze the synergistic effectiveness of RE policy tools. Based on the analysis, the following conclusions are drawn.

Provincial policy texts mainly use Targets & Planning and Industrial Development with lower stringency than the other three policy tools. Between 2005 and 2020, policy texts that relied on a single tool comprised the majority, averaging 66.8%. During the sample period, 46.8% and 54.2% of policy texts involved Targets & Planning and Industrial Development, respectively, significantly higher than the other three types, ranging from 5.9% to 22.5%. However, the tools used most frequently tend to be the least stringent. When assuming that the secondary policy texts are half as effective as primary policy texts, the average provincial policy stringency degrees of the five tools are 2.7858 (Operations Management), 2.233 (Price Regulation), 2.1822 (Financial Support), and 1.9912 (Targets & Planning), and 1.8254 (Industrial Development) respectively.

Issuing more stringent texts of command-and-control policies is more conducive to driving the RE progress than market-based policies. Our research findings affirm that heightened stringency levels of command-and-control policies, specifically Targets & Planning and Operations Management, are associated with increased per capita RE installed capacity. Conversely, the stringency of Industrial Development shows a negative relationship with RE development, while the effect of Financial Support is insignificant. Furthermore, Price Regulation policies exhibit a positive influence; however, it is noteworthy that they cannot be fully regarded as market-based tools, given that on-grid prices of RE in China are still dependent on government regulations.

RE policies focusing on Targets & Planning and Operations Management achieve effective synergy, while the combination of Financial Support and Industrial Development has the opposite effect. The synergistic effectiveness of policy tools depends on how they are combined. Association rule mining results show that in provincial policy texts of RE, Operations Management→Targets & Planning, Targets & Planning→Industrial Development, and Financial Support→Industrial Development have the strongest associations. Among them, Operations Management supports the more concrete implementation of abstract Targets & Planning policies, resulting in policy synergy and effective cooperation. Targets & planning can also mitigate the negative effects of Industrial Development policies in the short term. However, the simultaneous application of Financial Support and Industrial Development could lead to synergy failure. Financial Support policies further emphasize the priority of developing RE industries that give rise to the potential of excessive reliance on government assistance and "cutting off" competition between Top-tier companies and small-and-medium companies, competition between RE industry and traditional energy industries, exacerbating the market distortion and imbalanced industrial structure.

While secondary policy texts constitute a substantial portion of policy texts, the policy effects weaken as their importance levels relative to primary policy texts increase. During the sample period, primary policy texts accounted for only approximately 6.1% of the total policy texts, highlighting the critical significance of secondary policies. However, when the importance of secondary policy texts compared to the primary policy texts increases, the regression results indicate that the absolute value of the estimated main effect and interaction effect decreases. This could be attributed to the opportunity cost of policy resources resulting from the redundant publication of secondary policy texts with relatively low clarity and potential inconsistency in content due to interest conflicts with other administrative departments.

5.2. Policy implications

The following policy implications are proposed for Chinese governments based on the conclusions.

First, China's local governments should exercise prudence and deliberation in the design and issuance of future policy texts, particularly when incorporating industrial development and fiscal support simultaneously. Duplication of incentives from combining two market-based policy tools may result in wasted policy resources and elevated policy costs. Moreover, excessive government intervention and over-reliance on government support may diminish the effects of "innovative compensation" and exacerbate market distortions as well as welfare losses.

Second, intensity the stringency of targets & planning and operations management policies, and flexibly balance command-and-control measures with market-based instruments while taking into consideration the specific driving mechanisms. Combining targets & planning with operations management complements each other's strengths and creates positive synergies. However, placing over-reliance on mandatory measures not only consumes more government resources, but also curtails investment enthusiasm of nonstate-owned enterprises (Wu et al., 2024), hindering the marketization of RE. A reasonable mix of control and market-based policy tools, such as a combination of targets & planning and industrial development, can reduce regulatory costs and prevent excessive government intervention. However, it is crucial to strengthen the detail and rigor of targets & planning policies and reduce the compulsion of industrial development policies, to foster a more conducive and dynamic environment for renewable energy market players.

Finally, the pertinence of the RE policy text needs to be further enhanced. Secondary policy texts may not be able to play the primary policy's full role. If they play a more significant role than the primary policy, they bring redundancy, weaken the policy's effect, and hinder RE's efficiency. Thus, prioritizing the publishment of well-defined primary policy texts and streamlining secondary policy texts to reduce unnecessary complexity and stringency is crucial.

Credit author statement.

Xing Hu: Analysis, Writing- original draft, review, and editing.

Shiwei Yu: Conceptualization, Analysis, Writing- original draft, review and editing, Project administration.

Xu Fang: Data management, Programming, Writing- original draft.

Marten Ovaere: Conceptualization, Analysis, Writing- review and editing,

Acknowledgements

This study was supported by the National Natural Science Foundation of China (Nos. 72293572, 72174188, and 31961143006).

Appendix A. Appendixes

Appendix 1. Rating scale for policy text quantification

Table A1
The Rating Scale for the Strictness and Rigor of Local Policies.

Items	Scores	Criteria for policy strictness
	5	Local regulations promulgated by the local People's Congress and its Standing Committee
	4	Orders from the local government
	3	Normative documents of local government and local People's Congress and its Standing Committee
	2	Working documents of the local, provincial government and local People's Congress and its Standing Committee; Normative documents issued by other provincial administrative institutions
	1	Working documents issued by other provincial administrative institutions
Policy types	**Scores**	**Criteria for policy rigor**
Targets & Planning	1	Only refer to the general description of promoting the deployment and utilization of renewable energy sources or preparing renewable energy planning
	3	Set the expected target value of renewable energy production/consumption/output or other related indicators at a particular time point; Or demonstrate specific ways to expand the scale of renewable energy development and utilization, such as setting up demonstration pilots, building production bases, photovoltaic poverty alleviation power stations, etc.
	5	More details provided than the criteria for a score of three. For example, release a more detailed construction scale, allocation scheme or work scheme for the realization of the overall planning goals based on setting expected values at a certain point in time; Or issue a complete and detailed renewable energy development plan; Or organize the implementation and declaration of renewable energy projects in a specific way
Industrial Development	1	Only present an abstract expression of promoting the development of renewable energy and related equipment industry
	3	Propose specific ways or reasonable opinions to strengthen industrial cultivation, such as breakthrough key technologies, introducing professional and technical personnel, exhibition publicity of products and equipment, standard technical improvement, etc.
	5	More details provided than the criteria for a score of three. For example, organize and carry out relevant work and give notice according to the specific ways of industrial cultivation; Or develop a detailed guide or catalog for the key support areas in the RE industry
Financial Support	1	Only give basic information or description for renewable energy, such as increasing fiscal and tax support for renewable energy or perfecting incentive policies
	3	Determine specific forms, sources of funds, scopes, and objects of the support, such as providing financial discount interest for enterprises or industries that meet the standards. Or determine the intensity of fiscal and tax support, that is, specify the amount of funds, subsidy period, etc.
	5	More details provided than the criteria for a score of three. For example, organize and carry out the procedural work of the declaration, approval, and distribution of special funds based on clarifying the specific measures, scopes, and intensity of fiscal and tax support; Or release a complete catalog of awards or detailed management methods of special funds
Price Regulation	1	Only mention general contents of improving the renewable energy pricing mechanism or implementing the feed-in tariff policy
	3	Emphasize the concrete measures and rationalization suggestions to improve the price mechanism; Or specify the feed-in tariff price, implementation area, and implementation time for renewable energy.
	5	More details provided than the criteria for a score of three. For example, release formal measures of renewable energy electricity prices according to the specified feed-in tariff price or suggestions, Or formulate clear pricing mechanisms for renewable energy and electricity trading rules
Operations Management	1	Only mention general requirements for standardizing the process of renewable energy development
	3	Put forward reasonable requirements or specific opinions, such as organization, overall coordination, and improvement of service system, for the renewable energy development and grid-connection process or the production, sales, and utilization process of renewable energy products and their corresponding infrastructures
	5	More details provided than the criteria for a score of three. For example, aiming at proposed requirements or opinions, organize the implementation process of vetting, approval, records, supervision, and assessment of planned renewable energy projects construction; Or formulate detailed and systematic management measures regarding investment, enterprise operation, supervision, and regulation, etc.

Note: "Other provincial administrative institutions" means other institutions except provincial people's governments, provincial people's congresses, and Standing committees. If multiple issuing departments are involved, the highest score of policy strictness will be taken.

Appendix 2. Apriori algorithm

The present study uses an association rule learning solution to discover the internal relationship between different renewable energy policy tools in the database. The Apriori algorithm performs association rule mining, using candidate item sets and iterative layer-by-layer searches to find frequent item sets. The specific implementation process is shown below.

(1) The input database M with minimum support is scanned through the database M in a single trip. And the support of each 1-item set is calculated to obtain the set of frequent 1-item sets L_1.

(2) Connection: To generate a frequent k-item set L_k, a frequent candidate k-item set C_k is generated in advance. The set C_k of candidate frequent item sets is obtained by a connection operation with itself. Let l_1 and l_2 be the set of items in which let $l_i[j]$ denote the j th item. Assume that the items in the set are sorted in dictionary order and the first (k-2) items are the same, and only the last item is different; perform the connection:$(l_1[1] = l_2[1])\,(l_1[2] = l_2[2]) \wedge \ldots \wedge (l_1[k-2] = l_2[k-2]) \wedge (l_1[k-1] < l_2[k-1])$. The result of connecting l_1 and l_2 is $l_1[1]l_1[2]\cdots l_1[k-1]l_2[k-1]$. L_1 is used to find the set L_2 of frequent 2-item sets, L_2 is used to find L_3, and so on, until no frequent k-item sets can be found.

(3) Pruning: Since any infrequent (k-1)-item set cannot be a subset of the frequent k-item set, pruning is performed to find the frequent k-item set.

The above steps are repeated through an iterative loop until no new set of frequent itemsets can be generated.

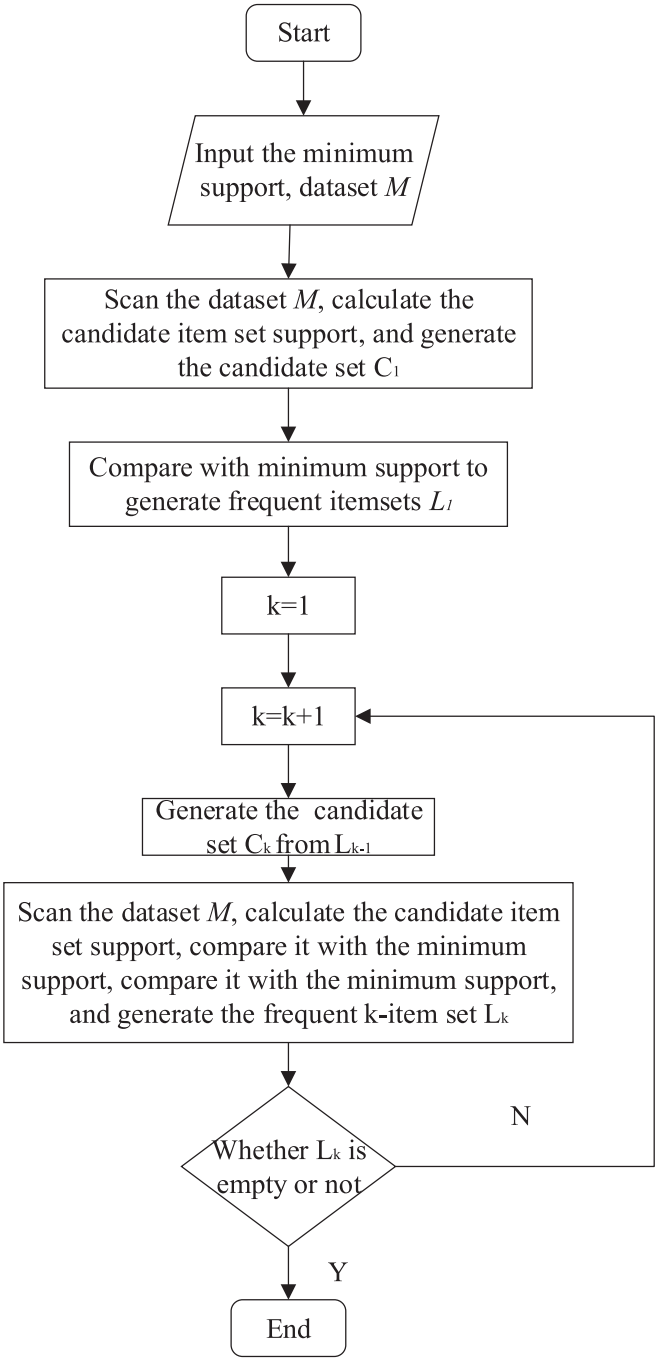


Fig. A1. The work process of the Apriori algorithm

Appendix 3. Descriptive statistics analysis

Table A2
Summary statistics.

Variable	Obs.	Mean	Std. Dev.	Min	Max	Unit
RED	480	0.185	0.451	0	4.121	kW/person
PGDP	480	4.278	2.541	0.681	14.86	10^4 yuan/person
ES	480	63.841	19.57	1.5	98.511	%
TI	480	0.087	0.168	0.001	1.469	–
$w = 0.25$						
TP	480	1.137	0.393	0	5	

(*continued on next page*)

Table A2 (*continued*)

Variable	Obs.	Mean	Std. Dev.	Min	Max	Unit
ID	480	1.037	0.442	0	5	
FS	480	1.482	0.95	0	9	
PR	480	1.44	1.099	0	6.875	
OM	480	1.906	1.506	0	25	
$w = 0.50$						
TP	480	1.991	0.5	0	5	
ID	480	1.825	0.545	0	5.125	
FS	480	2.182	1.004	0	9	
PR	480	2.233	1.305	0	8.75	
OM	480	2.786	1.558	0	25	
$w = 0.75$						
TP	480	2.846	0.644	0	7.125	
ID	480	2.614	0.691	0	5.813	
FS	480	2.883	1.141	0	9	
PR	480	3.026	1.609	0	10.625	
OM	480	3.666	1.706	0	25	
$w = 1$						
TP	480	3.715	0.791	0	9.5	
ID	480	3.409	0.851	0	6.5	
FS	480	3.6	1.33	0	9	
PR	480	3.821	1.968	0	12.5	
OM	480	4.576	1.922	0	25	

Appendix 4. Results of preliminary tests

Table A3
Results of cross-sectional dependence tests and unit root test.

Variable	CD test	CIPS test		Variable	CD test	CIPS test	
		Level	1st difference			Level	1st difference
ln*RED*	67.15***	−3.007***	−4.330***	ln*ES*	30.78***	−1.011	−3.345***
ln*PGDP*	83.03***	−0.390	−1.597*	ln*TI*	82.78***	−1.524*	−2.119***
$w = 0.25$				$w = 0.75$			
ln*TP*	23.64***	−2.263***	−4.839***	ln*TP*	7.65***	−2.769***	−4.644***
ln*ID*	26.56***	−2.599***	−4.845***	ln*ID*	20.60***	−3.191***	−4.803***
ln*FS*	30.43***	−3.740***	−5.250***	ln*FS*	32.07***	−3.516***	−5.221***
ln*PR*	42.08***	−4.610***	−5.236***	ln*PR*	38.80***	−4.320***	−5.389***
ln*OM*	20.07***	−3.558***	−4.649***	ln*OM*	13.30***	−3.934***	−4.919***
$w = 0.50$				$w = 1$			
ln*TP*	13.08***	−2.785***	−4.846***	ln*TP*	5.91***	−2.195***	−4.488***
ln*ID*	22.51***	−3.224***	−4.806***	ln*ID*	19.09***	−3.213***	−4.952***
ln*FS*	29.97***	−3.593***	−5.224***	ln*FS*	32.87***	−3.491***	−5.214***
ln*PR*	40.06***	−4.376***	−5.359***	ln*PR*	37.56***	−4.318***	−5.389***
ln*OM*	14.40***	−3.790***	−4.850***	ln*OM*	13.78***	−3.933***	−4.952***

Note: The critical values of the CIPS test are −1.5, −1.61, and − 1.8% at 10%, 5%, and 1% significance levels, respectively. Constant and time trends are not included.

Table A4
Results of Kao test for panel Cointegration.

Statistics	*Model* 1 ($w = 0.25$)	*Model* 2 ($w = 0.50$)	*Model* 3 ($w = 0.75$)	*Model 4* ($w = 1$)
Modified Dickey-Fuller t	−6.2486***	−6.1881***	−6.1805***	−6.1594***
Dickey-Fuller t	−8.8901***	−8.9084***	−8.8917***	−8.8822***
Augmented Dickey-Fuller t	−6.8622***	−6.9436***	−7.0871***	−7.2458***
Unadjusted Modified Dickey-Fuller t	−7.7057***	−7.7660***	−7.7691***	−7.8293***
Unadjusted Dickey-Fuller t	−9.3232***	−9.3820***	−9.3689***	−9.3864***

Note: ***, **, and * represent statistical significance at the level of 1%, 5%, and 10%, respectively.

Appendix 5. Results of robust tests

Table A5
Robust test results of changing measurement for renewable energy development.

		$l = 1$				$l = 2$		
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
	$w = 0.25$	$w = 0.50$	$w = 0.75$	$w = 1$	$w = 0.25$	$w = 0.50$	$w = 0.75$	$w = 1$
ln$TP_{t\text{-}l}$	0.577***	0.561***	0.548***	0.533***	0.640***	0.627***	0.614***	0.599***
	(0.126)	(0.112)	(0.105)	(0.102)	(0.039)	(0.043)	(0.045)	(0.045)
ln$ID_{t\text{-}l}$	−0.375***	−0.322***	−0.300***	−0.288***	−0.292***	−0.261***	−0.249***	−0.244***

(*continued on next page*)

Table A5 (*continued*)

		$l=1$				$l=2$		
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
	$w=0.25$	$w=0.50$	$w=0.75$	$w=1$	$w=0.25$	$w=0.50$	$w=0.75$	$w=1$
	(0.051)	(0.049)	(0.048)	(0.047)	(0.059)	(0.058)	(0.058)	(0.058)
$\ln FS_{t-l}$	−0.015	0.003	0.011	0.015	0.025	0.026	0.025	0.023
	(0.042)	(0.042)	(0.042)	(0.042)	(0.046)	(0.048)	(0.048)	(0.048)
$\ln PR_{t-l}$	0.185**	0.184**	0.183**	0.183**	0.076	0.073	0.072	0.072
	(0.075)	(0.074)	(0.073)	(0.072)	(0.046)	(0.045)	(0.045)	(0.044)
$\ln OM_{t-l}$	0.222***	0.196***	0.181***	0.170***	0.212***	0.188***	0.174***	0.164***
	(0.073)	(0.064)	(0.060)	(0.058)	(0.048)	(0.041)	(0.039)	(0.038)
$(\ln TP\times\ln OM)_{t-l}$	0.028**	0.028**	0.028***	0.027***	0.027***	0.027***	0.027***	0.027***
	(0.011)	(0.010)	(0.010)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
$(\ln TP\times\ln ID)_{t-l}$	0.028**	0.027**	0.025**	0.025**	0.023*	0.022*	0.021*	0.020*
	(0.012)	(0.011)	(0.010)	(0.010)	(0.013)	(0.012)	(0.012)	(0.011)
$(\ln ID\times\ln FS)_{t-l}$	−0.024***	−0.020***	−0.019***	−0.018***	−0.001	0.000	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
N	450	450	450	450	420	420	420	420
F statistic	7.28e+06***	5.92e+06***	2.73e+13***	1.47e+09***	1.33e+07***	6.61e+06***	5.08e+06***	4.21e+06***
R^2	0.593	0.593	0.593	0.594	0.570	0.571	0.571	0.571
Control	YES	YES	YES	YES	YES	YES	YES	YES
Individual effect	YES	YES	YES	YES	YES	YES	YES	YES
Time effect	YES	YES	YES	YES	YES	YES	YES	YES

Table A6
Robust test results of introducing lagged control variables.

		$l=1$				$l=2$		
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
	$w=0.25$	$w=0.50$	$w=0.75$	$w=1$	$w=0.25$	$w=0.50$	$w=0.75$	$w=1$
$\ln TP_{t-l}$	0.374***	0.361***	0.352***	0.342***	0.422***	0.408***	0.399***	0.388***
	(0.079)	(0.068)	(0.064)	(0.062)	(0.031)	(0.030)	(0.031)	(0.031)
$\ln ID_{t-l}$	−0.240***	−0.205***	−0.191***	−0.184***	−0.202***	−0.181***	−0.172***	−0.168***
	(0.035)	(0.034)	(0.034)	(0.033)	(0.040)	(0.038)	(0.038)	(0.038)
$\ln FS_{t-l}$	−0.010	0.001	0.005	0.008	0.017	0.018	0.017	0.016
	(0.025)	(0.024)	(0.025)	(0.025)	(0.021)	(0.023)	(0.024)	(0.024)
$\ln PR_{t-l}$	0.145***	0.144***	0.143***	0.142***	0.068**	0.066**	0.065**	0.064**
	(0.049)	(0.048)	(0.047)	(0.047)	(0.030)	(0.030)	(0.030)	(0.029)
$\ln OM_{t-l}$	0.137***	0.119***	0.109***	0.103***	0.123***	0.106***	0.097***	0.090***
	(0.045)	(0.040)	(0.038)	(0.036)	(0.032)	(0.028)	(0.027)	(0.026)
$(\ln TP\times\ln OM)_{t-l}$	0.018**	0.018**	0.018**	0.017***	0.018***	0.018***	0.018***	0.017***
	(0.007)	(0.007)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
$(\ln TP\times\ln ID)_{t-l}$	0.014*	0.013*	0.013*	0.012*	0.011	0.010	0.010	0.009
	(0.008)	(0.007)	(0.007)	(0.007)	(0.008)	(0.008)	(0.008)	(0.007)
$(\ln ID\times\ln FS)_{t-l}$	−0.014***	−0.012***	−0.011***	−0.010***	−0.002	−0.001	−0.001	−0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
$\ln PGDP_{t-1}$	13.168***	13.047***	12.983***	12.938***	14.090***	13.996***	13.944***	13.908***
	(3.410)	(3.427)	(3.435)	(3.437)	(4.389)	(4.442)	(4.469)	(4.485)
$\ln ES_{t-1}$	−4.034**	−3.994**	−3.976**	−3.961**	−3.414*	−3.395*	−3.386*	−3.377*
	(1.913)	(1.899)	(1.893)	(1.892)	(1.738)	(1.730)	(1.727)	(1.728)
$\ln TI_{t-1}$	1.434	1.466	1.479	1.481	2.413**	2.443***	2.457***	2.458***
	(1.099)	(1.079)	(1.069)	(1.065)	(0.885)	(0.874)	(0.868)	(0.867)
N	450	450	450	450	420	420	420	420
F statistic	1.02e+07***	2.87e+06***	8.20e+06***	2.29e+08***	1.91e+07***	6.46e+07***	2.99e+07***	1.40e+07***
R^2	0.652	0.652	0.653	0.653	0.631	0.631	0.631	0.631
Individual effect	YES	YES	YES	YES	YES	YES	YES	YES
Time effect	YES	YES	YES	YES	YES	YES	YES	YES

Table A7
Robust test results of changing sample period.

		$l=1$				$l=2$		
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
	$w=0.25$	$w=0.50$	$w=0.75$	$w=1$	$w=0.25$	$w=0.50$	$w=0.75$	$w=1$
$\ln TP_{t-l}$	0.349***	0.340***	0.332***	0.323***	0.397***	0.387***	0.379***	0.369***
	(0.074)	(0.067)	(0.063)	(0.061)	(0.024)	(0.028)	(0.030)	(0.031)
$\ln ID_{t-l}$	−0.247***	−0.212***	−0.197***	−0.189***	−0.195***	−0.175***	−0.166***	−0.162***

(*continued on next page*)

Table A7 (*continued*)

		$l = 1$				$l = 2$		
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
	$w = 0.25$	$w = 0.50$	$w = 0.75$	$w = 1$	$w = 0.25$	$w = 0.50$	$w = 0.75$	$w = 1$
	(0.034)	(0.034)	(0.034)	(0.033)	(0.041)	(0.041)	(0.042)	(0.042)
$\ln FS_{t-l}$	−0.006	0.005	0.009	0.011	0.021	0.021	0.020	0.019
	(0.021)	(0.021)	(0.021)	(0.021)	(0.021)	(0.023)	(0.024)	(0.024)
$\ln PR_{t-l}$	0.140***	0.139***	0.138***	0.137***	0.060*	0.058*	0.057*	0.057*
	(0.046)	(0.046)	(0.045)	(0.045)	(0.031)	(0.030)	(0.030)	(0.030)
$\ln OM_{t-l}$	0.129***	0.112***	0.103***	0.096**	0.125***	0.109***	0.100***	0.093***
	(0.044)	(0.039)	(0.036)	(0.035)	(0.033)	(0.029)	(0.028)	(0.027)
$(\ln TP \times \ln OM)_{t-l}$	0.018**	0.018**	0.017**	0.017**	0.018***	0.018***	0.017***	0.017***
	(0.007)	(0.007)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
$(\ln TP \times \ln ID)_{t-l}$	0.014*	0.013*	0.012*	0.012*	0.012	0.011	0.011	0.011
	(0.008)	(0.007)	(0.007)	(0.007)	(0.009)	(0.008)	(0.008)	(0.007)
$(\ln ID \times \ln FS)_{t-l}$	−0.014***	−0.012***	−0.011***	−0.011***	−0.001	0.000	0.000	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
N	420	420	420	420	390	390	390	390
F statistic	1.08e+06***	1.29e+05***	9.62e+05***	4.29e+08***	9.37e+05***	1.07e+06***	9.62e+05***	9.39e+05***
R^2	0.639	0.639	0.640	0.640	0.618	0.618	0.640	0.618
Control	YES	YES	YES	YES	YES	YES	YES	YES
Individual effect	YES	YES	YES	YES	YES	YES	YES	YES
Time effect	YES	YES	YES	YES	YES	YES	YES	YES

Table A8
Robust test results of centering for interaction terms.

		$l = 1$				$l = 2$		
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
	$w = 0.25$	$w = 0.50$	$w = 0.75$	$w = 1$	$w = 0.25$	$w = 0.50$	$w = 0.75$	$w = 1$
$\ln TP_{t-l}$	0.349***	0.355***	0.356***	0.354***	0.394***	0.399***	0.400***	0.397***
	(0.073)	(0.067)	(0.064)	(0.063)	(0.024)	(0.026)	(0.027)	(0.028)
$\ln ID_{t-l}$	−0.234***	−0.198***	−0.182***	−0.174***	−0.190***	−0.164***	−0.152***	−0.145***
	(0.033)	(0.032)	(0.031)	(0.030)	(0.037)	(0.036)	(0.036)	(0.035)
$\ln FS_{t-l}$	−0.002	0.001	0.002	0.002	0.019	0.019	0.019	0.018
	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)	(0.024)	(0.025)	(0.025)
$\ln PR_{t-l}$	0.141***	0.140***	0.139***	0.138***	0.060*	0.058*	0.058*	0.057*
	(0.046)	(0.045)	(0.045)	(0.044)	(0.030)	(0.030)	(0.029)	(0.029)
$\ln OM_{t-l}$	0.129***	0.122***	0.119***	0.117***	0.124***	0.119***	0.116***	0.114***
	(0.044)	(0.040)	(0.038)	(0.038)	(0.033)	(0.029)	(0.028)	(0.027)
$(\ln TP \times \ln OM)_{t-l}$	0.017**	0.017**	0.017***	0.017***	0.017***	0.017***	0.017***	0.017***
	(0.007)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.005)
$(\ln TP \times \ln ID)_{t-l}$	0.015*	0.014*	0.013*	0.013*	0.014	0.013	0.012	0.012
	(0.008)	(0.007)	(0.007)	(0.007)	(0.008)	(0.008)	(0.008)	(0.007)
$(\ln ID \times \ln FS)_{t-l}$	−0.014***	−0.012***	−0.011***	−0.011***	−0.001	0.000	0.000	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
N	450	450	450	450	420	420	420	420
F statistic	1.22e+06***	5.15e+06***	7.83e+06***	7.18e+06***	3.26e+06***	1.99e+06***	1.67e+06***	1.40e+06***
R^2	0.651	0.651	0.651	0.651	0.630	0.631	0.631	0.631
Control	YES	YES	YES	YES	YES	YES	YES	YES
Individual effect	YES	YES	YES	YES	YES	YES	YES	YES
Time effect	YES	YES	YES	YES	YES	YES	YES	YES

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eneco.2023.107104.

References

Abrell, J., Rausch, S., 2016. Cross-country electricity trade, renewable energy and European transmission infrastructure policy. J. Environ. Econ. Manag. 79, 87–113. https://doi.org/10.1016/j.jeem.2016.04.001.

Agrawal, R., Imielinski, T., Swami, A., 1993. Database mining: a performance perspective. IEEE Trans. Knowl. Data Eng. 5 (6), 914–925. https://doi.org/10.1109/69.250074.

Alam, M.M., Murad, M.W., 2020. The impacts of economic growth, trade openness and technological progress on renewable energy use in organization for economic co-operation and development countries. Renew. Energy 145, 382–390. https://doi.org/10.1016/j.renene.2019.06.054.

Alolo, M., Azevedo, A., El Kalak, I., 2020. The effect of the feed-in-system policy on renewable energy investments: Evidence from the EU countries. Energy Econ. 92, 104998. https://doi.org/10.1016/j.eneco.2020.104998.

Ambec, S., Cohen, M.A., Elgie, S., Lanoie, P., 2013. The porter hypothesis at 20: can environmental regulation enhance innovation and competitiveness? Rev. Environ. Econ. Policy 7 (1), 2–22. https://doi.org/10.1093/reep/res016.

Antonello, F., Baraldi, P., Shokry, A., Zio, E., Gentile, U., Serio, L., 2021. Association rules extraction for the identification of functional dependencies in complex technical infrastructures. Reliab. Eng. Syst. Saf. 209, 107305 https://doi.org/10.1016/j.ress.2020.107305.

Avraam, C., Bistline, J.E.T., Brown, M., Vaillancourt, K., Siddiqui, S., 2021. North American natural gas market and infrastructure developments under different

mechanisms of renewable policy coordination. Energy Policy 148, 111855. https://doi.org/10.1016/j.enpol.2020.111855.
Beiser-McGrath, L.F., Bernauer, T., Prakash, A., 2023. Command and control or market-based instruments? Public support for policies to address vehicular pollution in Beijing and New Delhi. Environ. Polit. 32 (4), 586–618. https://doi.org/10.1080/09644016.2022.2113608.
Bersalli, G., Menanteau, P., El-Methni, J., 2020. Renewable energy policy effectiveness: a panel data analysis across Europe and Latin America. Renew. Sust. Energ. Rev. 133, 110351 https://doi.org/10.1016/j.rser.2020.110351.
Bistline, J.E.T., Brown, M., Siddiqui, S.A., Vaillancourt, K., 2020. Electric sector impacts of renewable policy coordination: a multi-model study of the north American energy system. Energy Policy 145, 111707. https://doi.org/10.1016/j.enpol.2020.111707.
Botta, E., Koźluk, T., 2014. Measuring environmental policy stringency in OECD countries: a composite index approach. In: OECD Economics Department Working Papers, No. 1177.
Bourcet, C., 2020. Empirical determinants of renewable energy deployment: a systematic literature review. Energy Econ. 85, 104563 https://doi.org/10.1016/j.eneco.2019.104563.
BP, 2022. Statistcial Review of World Energy 2022, 71st edition. London.
Brunel, C., Levinson, A., 2016. Measuring the stringency of environmental regulations. Rev. Environ. Econ. Policy 10 (1), 47–67. https://doi.org/10.1093/reep/rev019.
Canzoneri, M.B., Gray, J.A., 1985. Monetary policy games and the consequences of non-cooperative behavior. Int. Econ. Rev. 26 (3), 547–564. https://doi.org/10.2307/2526702.
Canzoneri, M.B., Cumby, R.E., Diba, B.T., 2005. The need for international policy coordination: what's old, what's new, what's yet to come? J. Int. Econ. 66 (2), 363–384. https://doi.org/10.1016/j.jinteco.2004.03.005.
Carley, S., 2009. State renewable energy electricity policies: an empirical evaluation of effectiveness. Energy Policy 37 (8), 3071–3081. https://doi.org/10.1016/j.enpol.2009.03.062.
Carley, S., Miller, C.J., 2012. Regulatory stringency and policy drivers: a reassessment of renewable portfolio standards. Policy Stud. J. 40 (4), 730–756. https://doi.org/10.1111/j.1541-0072.2012.00471.x.
Carley, S., Davies, L.L., Spence, D.B., Zirogiannis, N., 2018. Empirical evaluation of the stringency and design of renewable portfolio standards. Nat. Energy 3 (9), 754–763. https://doi.org/10.1038/s41560-018-0202-4.
Carrillo, J.A., Mendoza, E.G., Nuguer, V., Roldán-Peña, J., 2021. Tight money-tight credit: coordination failure in the conduct of monetary and financial policies. Am. Econ. J. Macroecon. 13 (3), 37–73. https://doi.org/10.1257/mac.20180321.
Chong, Z., Wang, Q., Wang, L., 2023. Is the photovoltaic power generation policy effective in China? A quantitative analysis of policy synergy based on text mining. Technol. Forecast. Soc. Chang. 195, 122770 https://doi.org/10.1016/j.techfore.2023.122770.
Coggins, J.S., Swinton, J.R., 1996. The Price of pollution: a dual approach to valuing SO2Allowances. J. Environ. Econ. Manag. 30 (1), 58–72. https://doi.org/10.1006/jeem.1996.0005.
Costantini, V., Crespi, F., 2008. Environmental regulation and the export dynamics of energy technologies. Ecol. Econ. 66 (2–3), 447–460.
Costantini, V., Crespi, F., Martini, C., Pennacchio, L., 2015. Demand-pull and technology-push public support for eco-innovation: The case of the biofuels sector. Res. Policy 44 (3), 577–595.
Dehbozorgi, M.R., Rastegar, M., Sami, A., 2022. Data mining-based cause identification of momentary outages in power distribution systems. Sustain. Cities Soc. 77, 103587 https://doi.org/10.1016/j.scs.2021.103587.
del Río, P., 2014. On evaluating success in complex policy mixes: the case of renewable energy support schemes. Policy. Sci. 47 (3), 267–287. https://doi.org/10.1007/s11077-013-9189-7.
Derwent, 2023. Derwent Innovation Index.
Dong, C., Zhou, R., Li, J., 2021. Rushing for subsidies: the impact of feed-in tariffs on solar photovoltaic capacity development in China. Appl. Energy 281, 116007. https://doi.org/10.1016/j.apenergy.2020.116007.
Dong, C.G., 2012. Feed-in tariff vs. renewable portfolio standard: an empirical test of their relative effectiveness in promoting wind capacity development. Energy Policy 42, 476–485. https://doi.org/10.1016/j.enpol.2011.12.014.
Driscoll, J.C., Kraay, A.C., 1998. Consistent covariance matrix estimation with spatially dependent panel data. Rev. Econ. Stat. 80 (4), 549–560. https://doi.org/10.1162/003465398557825.
Du, Y., Takeuchi, K., 2020. Does a small difference make a difference? Impact of feed-in tariff on renewable power generation in China. Energy Econ. 87, 104710 https://doi.org/10.1016/j.eneco.2020.104710.
Eker, S., van Daalen, E., 2015. A model-based analysis of biomethane production in the Netherlands and the effectiveness of the subsidization policy under uncertainty. Energy Policy 82, 178–196. https://doi.org/10.1016/j.enpol.2015.03.019.
Fankhauser, S., Hepburn, C., Park, J., 2010. Combining multiple climate policy instruments: how not to do it. Clim. Change Econom. 1 (03), 209–225.
Friesenbichler, K.S., 2016. Policy interaction and the integration of volatile renewable energy. Environ. Econ. Policy Stud. 18 (2), 193–211. https://doi.org/10.1007/s10018-015-0130-5.
Galeotti, M., Salini, S., Verdolini, E., 2020. Measuring environmental policy stringency: approaches, validity, and impact on environmental innovation and energy efficiency. Energy Policy 136, 111052. https://doi.org/10.1016/j.enpol.2019.111052.
Gielen, D., Boshell, F., Saygin, D., Bazilian, M.D., Wagner, N., Gorini, R., 2019. The role of renewable energy in the global energy transformation. Energ. Strat. Rev. 24, 38–50.
Gunningham, N., Sinclair, D., 1999. Regulatory pluralism: designing policy mixes for environmental protection. Law & Pol. 21 (1), 49–76. https://doi.org/10.1111/1467-9930.00065.
Hamada, K., 1976. A strategic analysis of monetary interdependence. J. Polit. Econ. 84 (4, Part 1), 677–700. https://doi.org/10.1086/260471.
Han, J., Fu, Y., 1999. Mining multiple-level association rules in large databases. IEEE Trans. Knowl. Data Eng. 11 (5), 798–805. https://doi.org/10.1109/69.806937.
Hille, E., 2018. Pollution havens: international empirical evidence using a shadow price measure of climate policy stringency. Empir. Econ. 54 (3), 1137–1171. https://doi.org/10.1007/s00181-017-1244-3.
Hille, E., Shahbaz, M., 2019. Sources of emission reductions: market and policy-stringency effects. Energy Econ. 78, 29–43. https://doi.org/10.1016/j.eneco.2018.11.006.
Hoechle, D., 2007. Robust standard errors for panel regressions with cross-sectional dependence. Stata J. 7 (3), 281–312.
Hombach, L.E., Walther, G., 2015. Pareto-efficient legal regulation of the (bio)fuel market using a bi-objective optimization model. Eur. J. Oper. Res. 245 (1), 286–295. https://doi.org/10.1016/j.ejor.2015.02.039.
Hood, C.C., 1983. Exploring Government's toolshed. In: Hood, C.C. (Ed.), The Tools of Government. Macmillan Education UK, London, pp. 1–15.
Howlett, M., Ramesh, M., 1995. Studying Public Policy: Policy Cycles and Policy Subsystems. Oxford university press, Toronto.
Hu, X., Guo, Y., Zheng, Y., Liu, L.-C., Yu, S., 2022. Which types of policies better promote the development of renewable energy? Evidence from China's provincial data. Renew. Energy 198, 1373–1382. https://doi.org/10.1016/j.renene.2022.08.139.
Hughes, C.E., Ritter, A., Mabbitt, N., 2013. Drug policy coordination: identifying and assessing dimensions of coordination. Int. J. Drug Pol. 24 (3), 244–250. https://doi.org/10.1016/j.drugpo.2012.08.004.
Im, K.S., Pesaran, M.H., Shin, Y., 2003. Testing for unit roots in heterogeneous panels. J. Econ. 115 (1), 53–74. https://doi.org/10.1016/S0304-4076(03)00092-7.
IRENA, 2022. In: Dhabi, A. (Ed.), Renewable Energy Targets in 2022: A Guide to Design. International Renewable Energy Agency.
Jaffe, A.B., Stavins, R.N., 1995. Dynamic incentives of environmental regulations: the effects of alternative policy instruments on technology diffusion. J. Environ. Econ. Manag. 29 (3), S43–S63. https://doi.org/10.1006/jeem.1995.1060.
Kao, C., 1999. Spurious regression and residual-based tests for cointegration in panel data. J. Econ. 90 (1), 1–44. https://doi.org/10.1016/S0304-4076(98)00023-2.
Komor, P., Bazilian, M., 2005. Renewable energy policy goals, programs, and technologies. Energy Policy 33 (14), 1873–1881.
Kwon, T.-H., 2018. Policy synergy or conflict for renewable energy support: case of RPS and auction in South Korea. Energy Policy 123, 443–449. https://doi.org/10.1016/j.enpol.2018.09.016.
Lambin, E.F., Meyfroidt, P., Rueda, X., Blackman, A., Börner, J., Cerutti, P.O., Dietsch, T., Jungmann, L., Lamarque, P., Lister, J., Walker, N.F., Wunder, S., 2014. Effectiveness and synergies of policy instruments for land use governance in tropical regions. Glob. Environ. Chang. 28, 129–140. https://doi.org/10.1016/j.gloenvcha.2014.06.007.
Landry, R., Varone, F., 2005. The choice of policy instruments: confronting the deductive and the interactive approaches. In: Designing Government. From Instruments to Governance, pp. 106–131.
Lema, A., Ruby, K., 2007. Between fragmented authoritarianism and policy coordination: creating a Chinese market for wind energy. Energy Policy 35 (7), 3879–3890. https://doi.org/10.1016/j.enpol.2007.01.025.
Li, C., Xia, W., Wang, L., 2023. Synergies of green policies and their pollution reduction effects: quantitative analysis of China's green policy texts. J. Clean. Prod. 412, 137360 https://doi.org/10.1016/j.jclepro.2023.137360.
Li, J., Ye, S., 2021. Regional policy synergy and haze governance—empirical evidence from 281 prefecture-level cities in China. Environ. Sci. Pollut. Res. 28 (9), 10763–10779. https://doi.org/10.1007/s11356-020-11251-8.
Liao, Z., 2016. The evolution of wind energy policies in China (1995–2014): an analysis based on policy instruments. Renew. Sust. Energ. Rev. 56, 464–472. https://doi.org/10.1016/j.rser.2015.11.097.
Linares, P., Santos, F.J., Ventosa, M., 2008. Coordination of carbon reduction and renewable energy support policies. Clim. Pol. 8 (4), 377–394. https://doi.org/10.3763/cpol.2007.0361.
Liu, D., Wang, D., 2022. Evaluation of the synergy degree of industrial de-capacity policies based on text mining: a case study of China's coal industry. Res. Policy 76, 102547. https://doi.org/10.1016/j.resourpol.2021.102547.
Liu, D., Wang, D., Mao, J., 2023. Study on policy synergy strategy of the central government and local governments in the process of coal de-capacity: based on a two-stage evolutionary game method. Res. Policy 80, 103132. https://doi.org/10.1016/j.resourpol.2022.103132.
Liu, J., 2019. China's renewable energy law and policy: a critical review. Renew. Sust. Energ. Rev. 99, 212–219. https://doi.org/10.1016/j.rser.2018.10.007.
Liu, W., Zhang, X., Feng, S., 2019. Does renewable energy policy work? Evidence from a panel data analysis. Renew. Energy 135, 635–642. https://doi.org/10.1016/j.renene.2018.12.037.
Lu, W., Wu, H., Yang, S., Tu, Y., 2022. Effect of environmental regulation policy synergy on carbon emissions in China under consideration of the mediating role of industrial structure. J. Environ. Manag. 322, 116053 https://doi.org/10.1016/j.jenvman.2022.116053.
Marques, A.C., Fuinhas, J.A., Pires Manso, J.R., 2010. Motivations driving renewable energy in European countries: a panel data approach. Energy Policy 38 (11), 6877–6885. https://doi.org/10.1016/j.enpol.2010.07.003.
Martínez-Zarzoso, I., Bengochea-Morancho, A., Morales-Lage, R., 2019. Does environmental policy stringency foster innovation and productivity in OECD

countries? Energy Policy 134, 110982. https://doi.org/10.1016/j.enpol.2019.110982.
McDonnell, L.M., Elmore, R.F., 1987. Getting the job done: alternative policy instruments. Educ. Eval. Policy Anal. 9 (2), 133–152. https://doi.org/10.3102/01623737009002133.
Menz, F.C., Vachon, S., 2006. The effectiveness of different policy regimes for promoting wind power: experiences from the states. Energy Policy 34 (14), 1786–1796. https://doi.org/10.1016/j.enpol.2004.12.018.
Metcalfe, L., 1994. International policy co-ordination and public management reform. Int. Rev. Adm. Sci. 60 (2), 271–290. https://doi.org/10.1177/002085239406000208.
Mulder, M., Zeng, Y., 2018. Exploring interaction effects of climate policies: a model analysis of the power market. Resour. Energy Econ. 54, 165–185. https://doi.org/10.1016/j.reseneeco.2018.09.002.
NEA, 2022. Narional Energy Administration. China's Total Utilization Of Renewable Energy Reached 750 Million Metric Tons of Standard Coal in 2022. http://www.gov.cn/xinwen/2022-06/24/content_5697606.htm (accessed 15 October 2022).
NEA, 2023. Narional Energy Administration. Transcript of the National Energy Administration's Q1 2023 Press Conference. http://www.nea.gov.cn/2023-02/13/c_1310697149.htm accessed 22 February 2023.
Nilsson, M., Zamparutti, T., Petersen, J.E., Nykvist, B., Rudberg, P., McGuinn, J., 2012. Understanding policy coherence: analytical framework and examples of sector–environment policy interactions in the EU. Environ. Policy Gov. 22 (6), 395–423. https://doi.org/10.1002/eet.1589.
Ostry, J.D., Ghosh, A.R., 2016. On the obstacles to international policy coordination. J. Int. Money Financ. 67, 25–40. https://doi.org/10.1016/j.jimonfin.2015.06.008.
Oudiz, G., Sachs, J., 1985. International policy coordination in dynamic macroeconomic models. In: International Economic Policy Coordination. Cambridge University Press, pp. 274–330.
Palage, K., Lundmark, R., Söderholm, P., 2019. The innovation effects of renewable energy policies and their interaction: the case of solar photovoltaics. Environ. Econ. Policy Stud. 21 (2), 217–254. https://doi.org/10.1007/s10018-018-0228-7.
Pang, S., Dou, S., Li, H., 2020. Synergy effect of science and technology policies on innovation: evidence from China. PLoS One 15 (10), e0240515. https://doi.org/10.1371/journal.pone.0240515.
Pesaran, M.H., 2007. A simple panel unit root test in the presence of cross-section dependence. J. Appl. Econ. 22 (2), 265–312.
Pesaran, M.H., 2021. General diagnostic tests for cross-sectional dependence in panels. Empir. Econ. 60 (1), 13–50. https://doi.org/10.1007/s00181-020-01875-7.
Peters, B.G., 2015. Pursuing Horizontal Management: The Politics of Public Sector Coordination. University Press of Kansas.
Peters, B.G., 2018. The challenge of policy coordination. Pol. Des. Pract. 1 (1), 1–11. https://doi.org/10.1080/25741292.2018.1437946.
Popiolek, N., Thais, F., 2016. Multi-criteria analysis of innovation policies in favour of solar mobility in France by 2030. Energy Policy 97, 202–219. https://doi.org/10.1016/j.enpol.2016.07.036.
Popp, D., Santen, N., Fisher-Vandend, K., Webster, M., 2013. Technology variation vs. R&D uncertainty: what matters most for energy patent success? Resour. Energy Econ. 35 (4), 505–533. https://doi.org/10.1016/j.reseneeco.2013.05.002.
Ren, Q., Albrecht, J., 2023. Toward circular economy: The impact of policy instruments on circular economy innovation for European small medium enterprises. Ecol. Econ. 207, 107761. https://doi.org/10.1016/j.ecolecon.2023.107761.
Robinson, C., Schumacker, R.E., 2009. Interaction effects: centering, variance inflation factor, and interpretation issues. Multiple Linear Regress. Viewpoints 35 (1), 6–11.
Rodrik, D., 2015. Green industrial policy. Oxf. Rev. Econ. Policy 30 (3), 469–491. https://doi.org/10.1093/oxrep/gru025.
Salamon, L.M., 2002. The Tools of Government: A Guide to the New Governance. Oxford University Press, New York.
Saquete, E., Zubcoff, J., Gutiérrez, Y., Martínez-Barco, P., Fernández, J., 2022. Why are some social-media contents more popular than others? Opinion and association rules mining applied to virality patterns discovery. Expert Syst. Appl. 197, 116676 https://doi.org/10.1016/j.eswa.2022.116676.
Sato, M., Dechezleprêtre, A., 2015. Asymmetric industrial energy prices and international trade. Energy Econ. 52, S130–S141. https://doi.org/10.1016/j.eneco.2015.08.020.
Saussay, A., Zugravu-Soilita, N., 2023. International production chains and the pollution offshoring hypothesis: an empirical investigation. Resour. Energy Econ. 101357 https://doi.org/10.1016/j.reseneeco.2023.101357.
Schmidt, J., Lehecka, G., Gass, V., Schmid, E., 2013. Where the wind blows: assessing the effect of fixed and premium based feed-in tariffs on the spatial diversification of wind turbines. Energy Econ. 40, 269–276. https://doi.org/10.1016/j.eneco.2013.07.004.
Schneider, A., Ingram, H., 1990. Behavioral assumptions of policy tools. J. Polit. 52 (2), 510–529. https://doi.org/10.2307/2131904.
Shang, H., Lu, D., Zhou, Q., 2021. Early warning of enterprise finance risk of big data mining in internet of things based on fuzzy association rules. Neural Comput. & Applic. 33 (9), 3901–3909. https://doi.org/10.1007/s00521-020-05510-5.
Shrestha, S., Dhakal, S., 2019. An assessment of potential synergies and trade-offs between climate mitigation and adaptation policies of Nepal. J. Environ. Manag. 235, 535–545. https://doi.org/10.1016/j.jenvman.2019.01.035.
Sovacool, B.K., 2010. A comparative analysis of renewable electricity support mechanisms for Southeast Asia. Energy 35 (4), 1779–1793. https://doi.org/10.1016/j.energy.2009.12.030.
Spillias, S., Kareiva, P., Ruckelshaus, M., McDonald-Madden, E., 2020. Renewable energy targets may undermine their sustainability. Nat. Clim. Chang. 10 (11), 974–976. https://doi.org/10.1038/s41558-020-00939-x.
Sun, X., Dong, Y., Wang, Y., Ren, J., 2022. Sources of greenhouse gas emission reductions in OECD countries: composition or technique effects. Ecol. Econ. 193, 107288.
Tandan, M., Acharya, Y., Pokharel, S., Timilsina, M., 2021. Discovering symptom patterns of COVID-19 patients using association rule mining. Comput. Biol. Med. 131, 104249 https://doi.org/10.1016/j.compbiomed.2021.104249.
Tang, M., Li, X., Zhang, Y., Wu, Y., Wu, B., 2020. From command-and-control to market-based environmental policies: optimal transition timing and China's heterogeneous environmental effectiveness. Econ. Model. 90, 1–10. https://doi.org/10.1016/j.econmod.2020.04.021.
Toivonen, H., 1996. Sampling large databases for association rules. Vldb 134–145.
Tu, Q., Mo, J.-L., 2017. Coordinating carbon pricing policy and renewable energy policy with a case study in China. Comput. Ind. Eng. 113, 294–304. https://doi.org/10.1016/j.cie.2017.09.026.
Tuladhar, S.D., Mankowski, S., Bernstein, P., 2014. Interaction effects of market-based and command-and-control policies. Energy J. 35 (Special Issue).
Ünvan, Y.A., 2021. Market basket analysis with association rules. Communicat. Stat. Theory Methods 50 (7), 1615–1628. https://doi.org/10.1080/03610926.2020.1716255.
van den Bergh, J., Castro, J., Drews, S., Exadaktylos, F., Foramitti, J., Klein, F., Konc, T., Savin, I., 2021. Designing an effective climate-policy mix: accounting for instrument synergy. Clim. Pol. 21 (6), 745–764. https://doi.org/10.1080/14693062.2021.1907276.
Vedung, E., 1998. Policy instruments: typologies and theories. In: Bemelmans-Videc, M.-L., Rist, R.C., Vedung, E. (Eds.), Carrots, Sticks and Sermons, 1st Edition ed. Routledge, p. 38.
Viguié, V., Hallegatte, S., 2012. Trade-offs and synergies in urban climate policies. Nat. Clim. Chang. 2 (5), 334–337. https://doi.org/10.1038/nclimate1434.
Wang, F., Li, K., Duić, N., Mi, Z., Hodge, B.-M., Shafie-khah, M., Catalão, J.P.S., 2018. Association rule mining based quantitative analysis approach of household characteristics impacts on residential electricity consumption patterns. Energy Convers. Manag. 171, 839–854. https://doi.org/10.1016/j.enconman.2018.06.017.
Wu, H., Zhong, R., Wang, Z., Qu, Y., Yang, X., Hao, Y., 2024. How does industrial intellectualization affect energy intensity? Evidence from China. Energy J. 45 (2) https://doi.org/10.5547/01956574.45.2.hawu.
Yan, Z., Zhou, Z., Du, K., 2023. How does environmental regulatory stringency affect energy consumption? Evidence from Chinese firms. Energy Econ. 118, 106503 https://doi.org/10.1016/j.eneco.2023.106503.
Yu, S., Lu, T., Hu, X., Liu, L., Wei, Y.-M., 2021. Determinants of overcapacity in China's renewable energy industry: evidence from wind, photovoltaic, and biomass energy enterprises. Energy Econ. 97, 105056 https://doi.org/10.1016/j.eneco.2020.105056.
Yin, H., Powers, N., 2010. Do state renewable portfolio standards promote in-state renewable generation? Energy Policy 38 (2), 1140–1149. doi:10.1016/j.enpol.2009.10.067.
Yu, S., Hu, X., Liu, J., 2022. Determinants of wind power curtailment in China: evidence from provincial panel data. Appl. Econ. 1-14 https://doi.org/10.1080/00036846.2022.2130146.
Zha, D., Jiang, P., Zhang, C., Xia, D., Cao, Y., 2023. Positive synergy or negative synergy: an assessment of the carbon emission reduction effect of renewable energy policy mixes on China's power sector. Energy Policy 183, 113782.
Zhang, G., Gao, Y., Li, J., Su, B., Chen, Z., Lin, W., 2022. China's environmental policy intensity for 1978–2019. Sci. Data 9 (1), 75.
Zhang, J., Liu, M., Zhang, B., 2023a. Environmental policy integration and performance: the effectiveness of China's one permit management reform. Environ. Impact Assess. Rev. 98, 106951 https://doi.org/10.1016/j.eiar.2022.106951.
Zhang, P., Zhou, D., Guo, J., 2023b. Policy complementary or policy crowding-out? Effects of cross-instrumental policy mix on green innovation in China. Technol. Forecast. Soc. Chang. 192, 122530.
Zhang, S., Andrews-Speed, P., Zhao, X., He, Y., 2013. Interactions between renewable energy policy and renewable energy industrial policy: a critical analysis of China's policy approach to renewable energies. Energy Policy 62, 342–353.
Zhao, X., Li, S., Zhang, S., Yang, R., Liu, S., 2016. The effectiveness of China's wind power policy: an empirical analysis. Energy Policy 95, 269–279. https://doi.org/10.1016/j.enpol.2016.04.050.